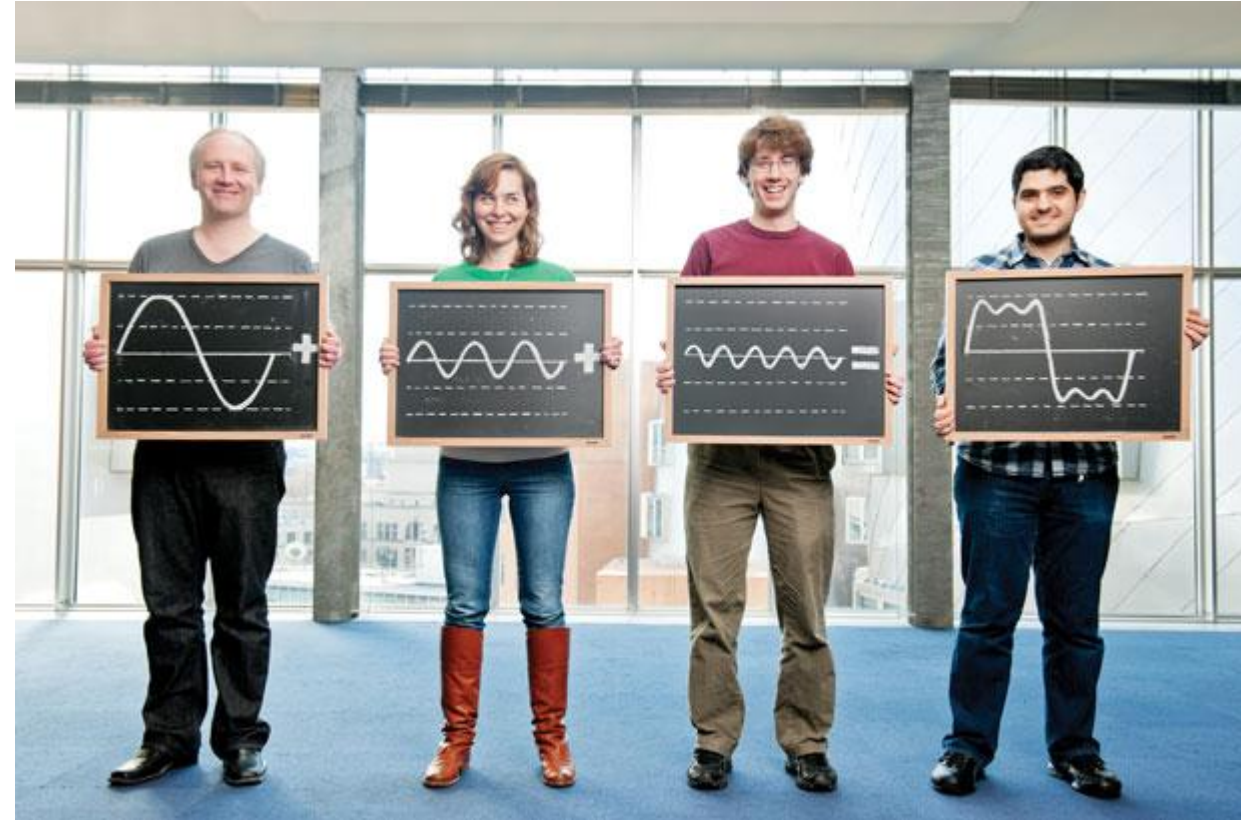




AY2026 MA05

Fourier Analysis

フーリエ解析



Class Information

Lectures (10+4): Period 7, 8, Monday (月曜日), Thursday (木曜日)



Grades:

Lecture Attendance $> 2/3$

30% Assignments (Submission Attendance $> 2/3$)

6% Quiz

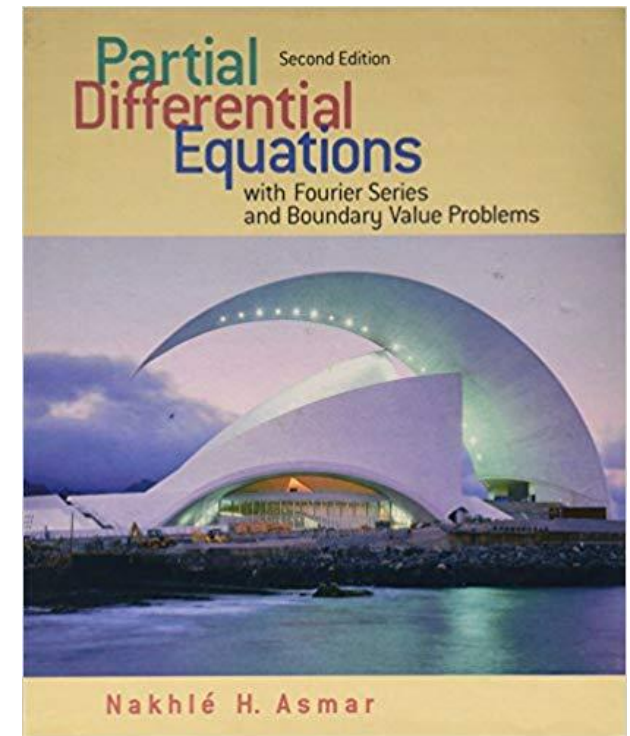
64% Final Examination (No re-exam)

+3 Bonus Points

Textbook: Partial Differential Equations with Fourier Series

and Boundary Value Problems, 2nd/3rd,

Nakhlé H. Asmar, University of Missouri, USA



Class Materials

Lecture Slides & Assignments

<https://web-ext.u-aizu.ac.jp/~xiangli/teaching/MA05/index.html>



Scan Me

Prerequisites 先修科目

MA01 Linear Algebra (線形代数 I)

MA03 Calculus I (微積分 I) and MA04 Calculus II (微積分 II)

Important related courses 重要な関連コース:

MA06 Complex Analysis (複数関数論)

IT03 Image Processing (画像処理論)

IT08 Signal Processing and Linear System (信号処理と線形システム)

IT09 Sound and Audio Processing (音響音声処理論)

What we will cover

Full syllabus on course website

Chapter 2

- 1: Part1. **Fourier series expansion** (Orthogonal system of the function space)
- 2: Part1. **Fourier series expansion** (Fourier series of trigonometric functions)
- 3: Part1. **Fourier series expansion (Exercise)**
- 4: Part2. **Properties of Fourier series** (Convergence condition of Fourier series)
- 5: Part2. **Properties of Fourier series** (Parseval's theorem, Weierstrass' theorem)
- 6: Part2. **Properties of Fourier series (Exercise)**

Chapter 7

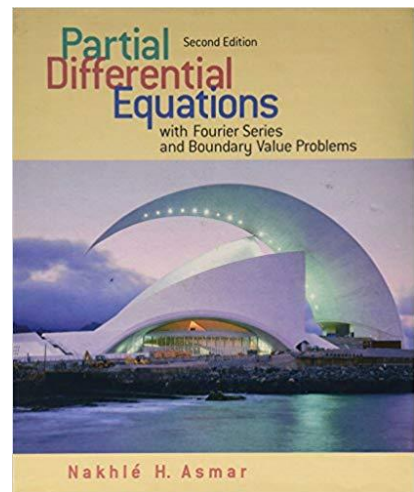
- 7: Part3. **Fourier integral** (Introduction from Fourier series, Fourier transform)
- 8: Part3. **Fourier integral** (Parseval's theorem, convolution)
- 9: Part3. **Fourier integral (Exercise)**

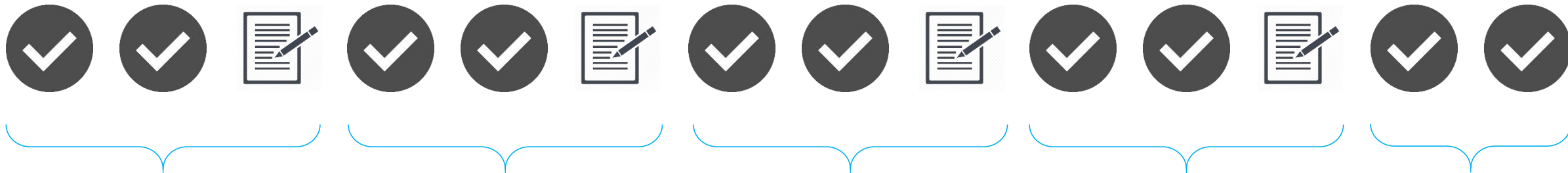
Chapter 8

- 10: Part4. **Laplace transform** (Introduction from Fourier transform)
- 11: Part4. **Laplace transform** (Ordinary differential equations of constant coefficients)
- 12: Part4. **Laplace transform (Exercise)**

Chapter 10

- 13: Part5. **Discrete Fourier transform** (Introduction from Fourier series)
- 14: Part5. **Discrete Fourier transform** (FFT(Fast Fourier Transform))





Fourier Series Expansion

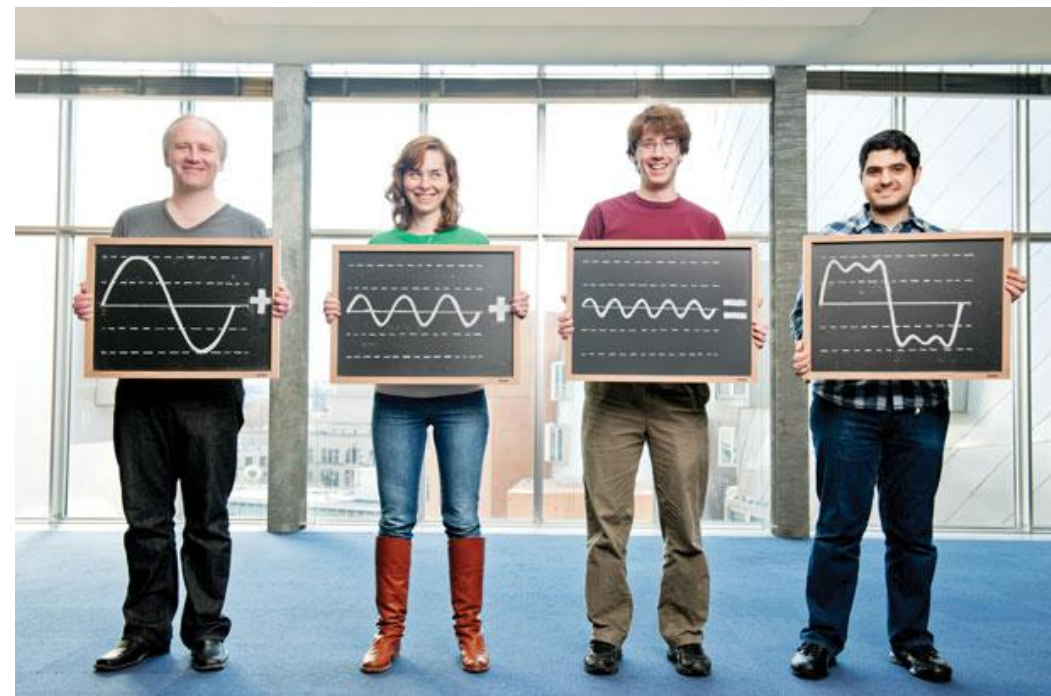
Fourier Integral

Laplace Transform

Properties of Fourier Series

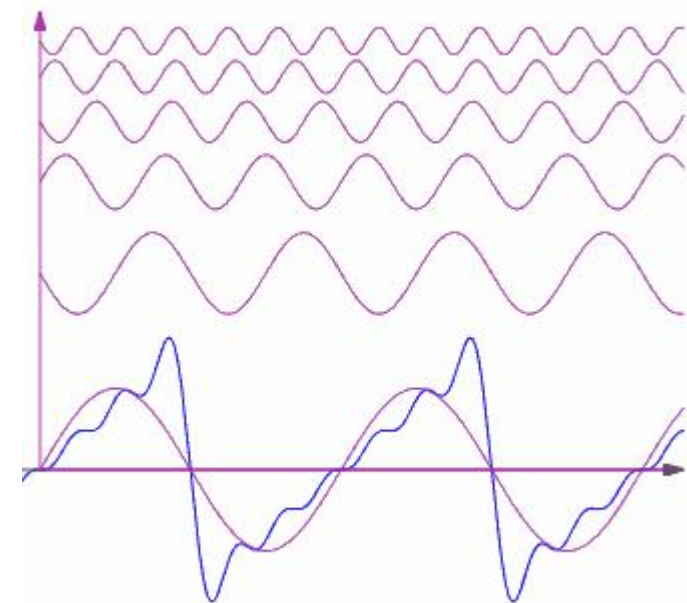
Discrete Fourier Transform

Fourier Analysis





Lecture 1



Orthogonal System and Fourier Series

直交システム と フーリエ級数

Outline of Lecture 1

1.1 Introduction of Fourier Analysis

フーリエ解析の紹介

1.2 Periodic Functions

周期関数

1.3 The Trigonometric System and Orthogonality

三角法システムと直交性

1.4 Fourier Series

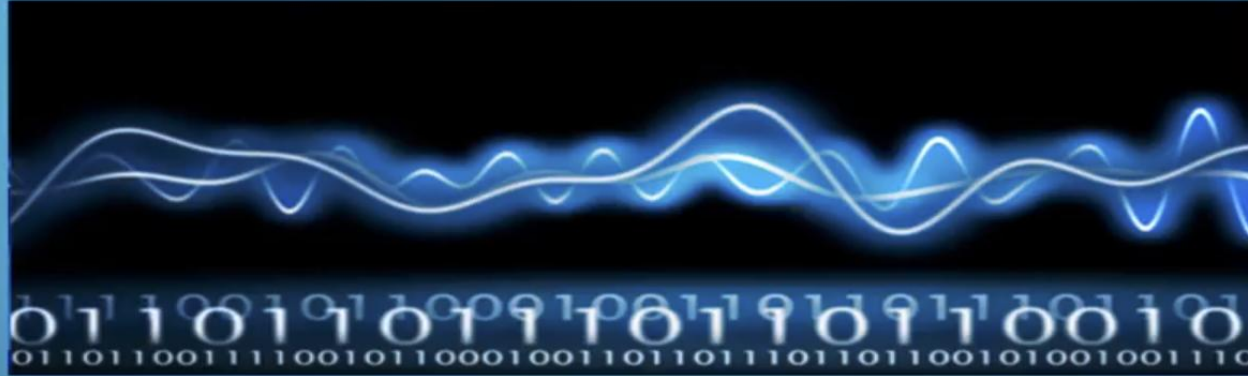
フーリエ級数

1.1 Introduction of Fourier Analysis

フーリエ解析の紹介

1.1 Introduction of Fourier Analysis (フーリエ解析の紹介)

WHAT IS SIGNAL ?



"A SIGNAL IS A SOURCE OF INFORMATION, GENERALLY A PHYSICAL QUANTITY, WHICH VARIES WITH RESPECT TO TIME, SPACE, TEMPERATURE OR LIKE ANY OTHER INDEPENDENT VARIABLE"

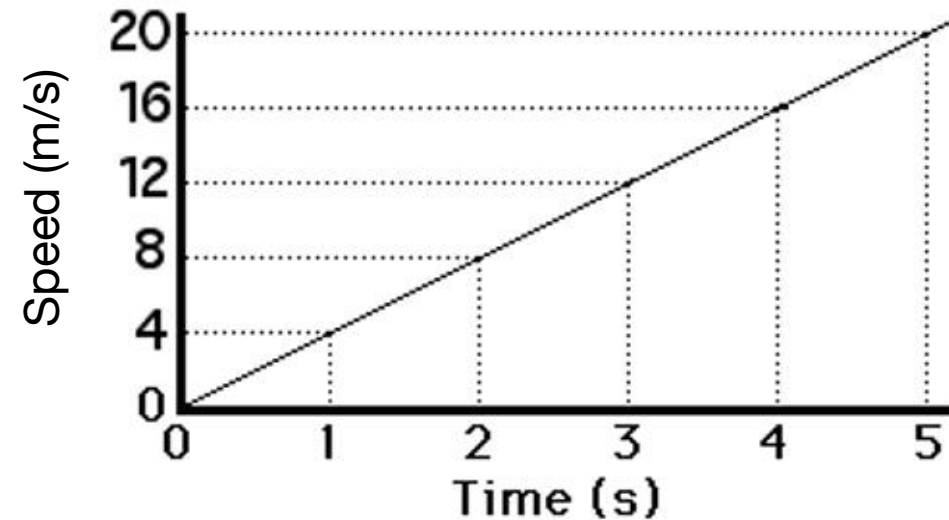
(翻訳) 信号 (英: Signal) は情報のソースであり、一般的には物理量であり、時間、空間、温度などの独立した変数によって変化します。

1.1 Introduction of Fourier Analysis (フーリエ解析の紹介)

Signals usually means physical quantities (**amplitude, dependent variable**) that change as a function of time, space, or some other **independent variable**.



Dependent Variable (Y)



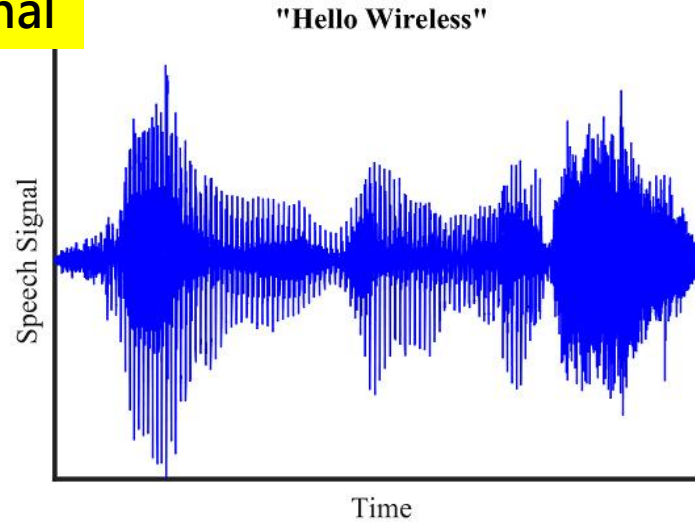
Independent Variable (X)

Any signal whose samples are a single-valued function of one independent variable is referred to as a **scalar one-dimensional signal** (simply as a **signal**).

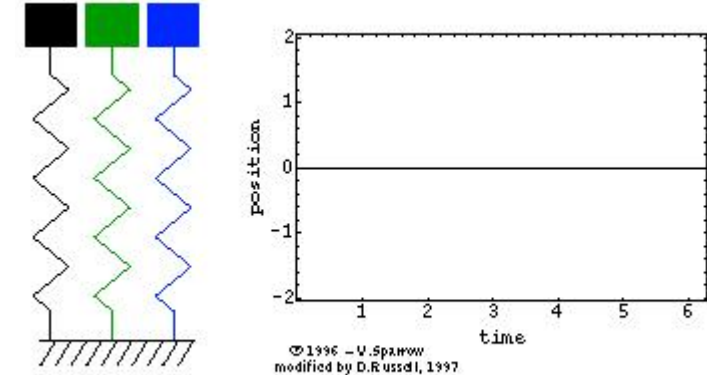
1.1 Introduction of Fourier Analysis (フーリエ解析の紹介)

Signals in real-world 現実世界の信号

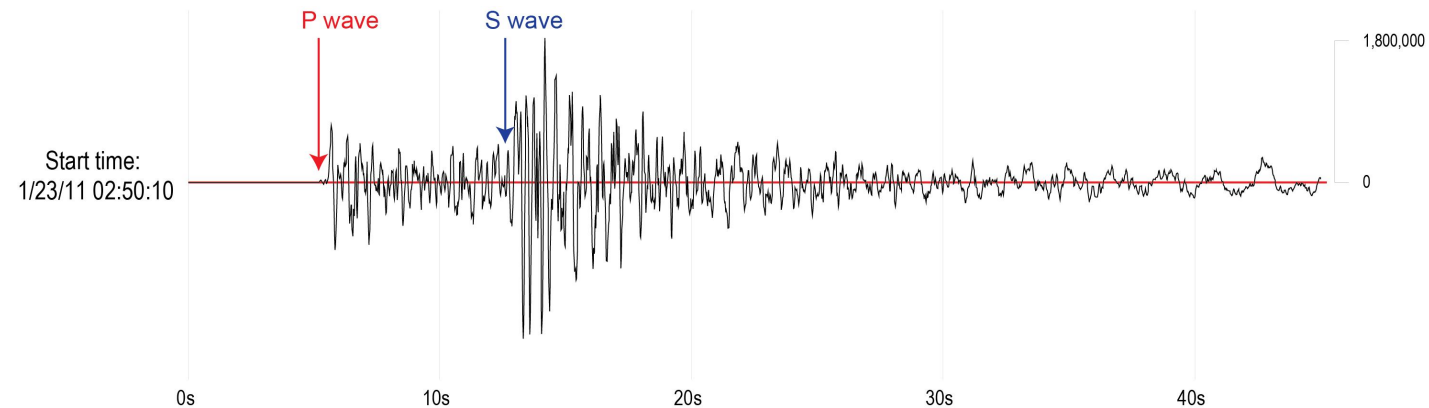
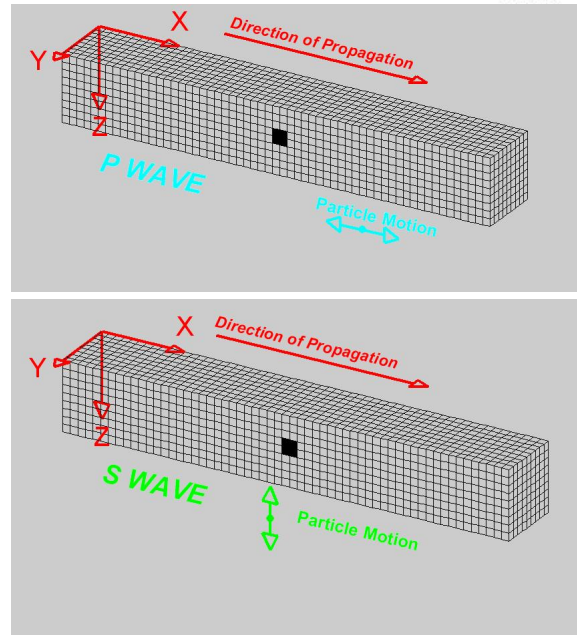
One-Dimensional (1D) Signal



Mechanical Vibrations



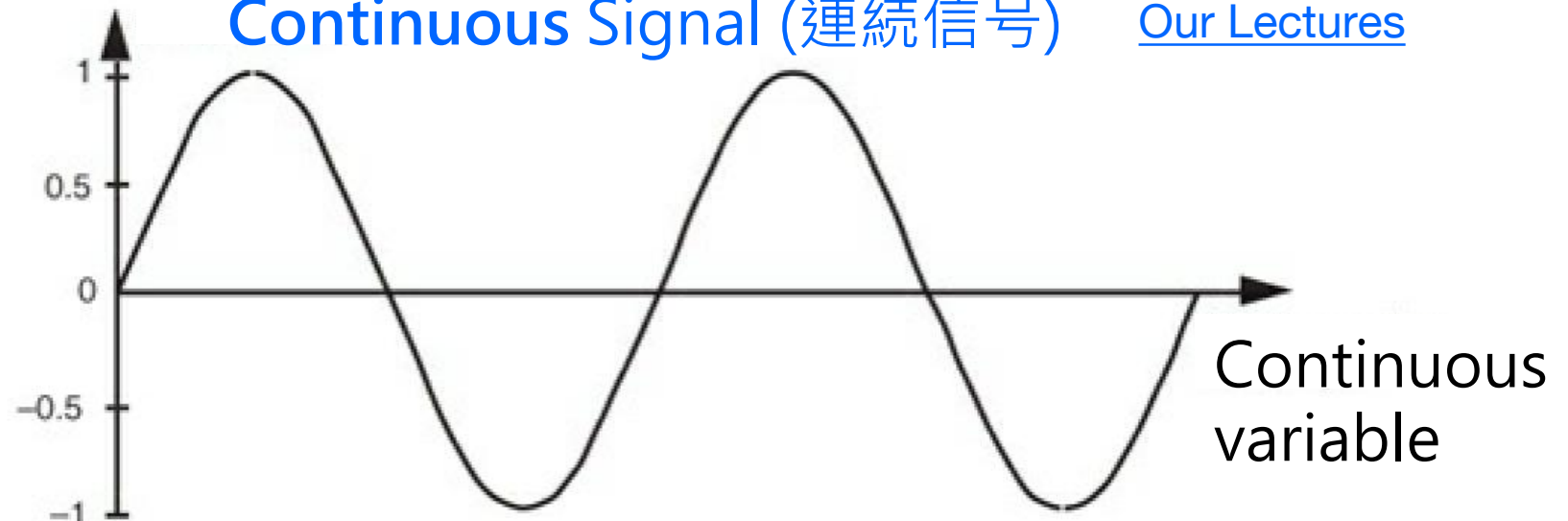
Earthquake Signal 地震信号



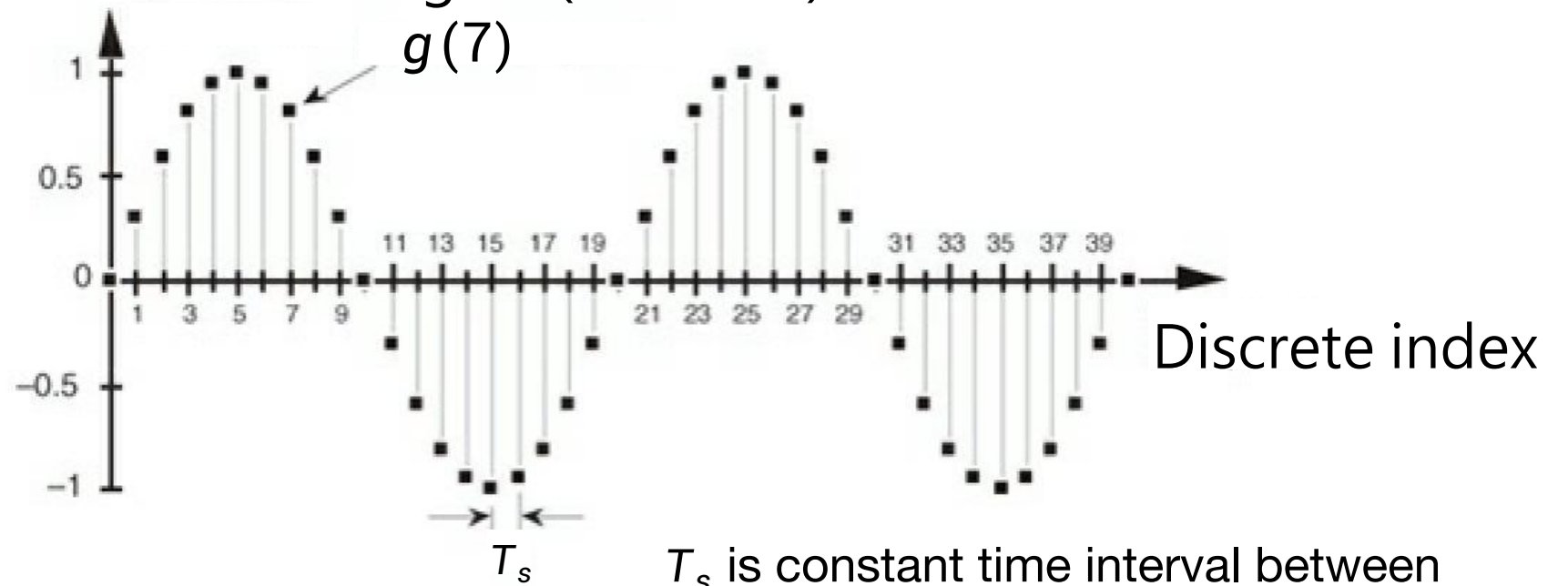
Types of Signal (1D)

Continuous Signal (連続信号)

[Our Lectures](#)



Discrete Signal (離散信号)



T_s is constant time interval between samples.

Two-Dimensional (2D) Signal

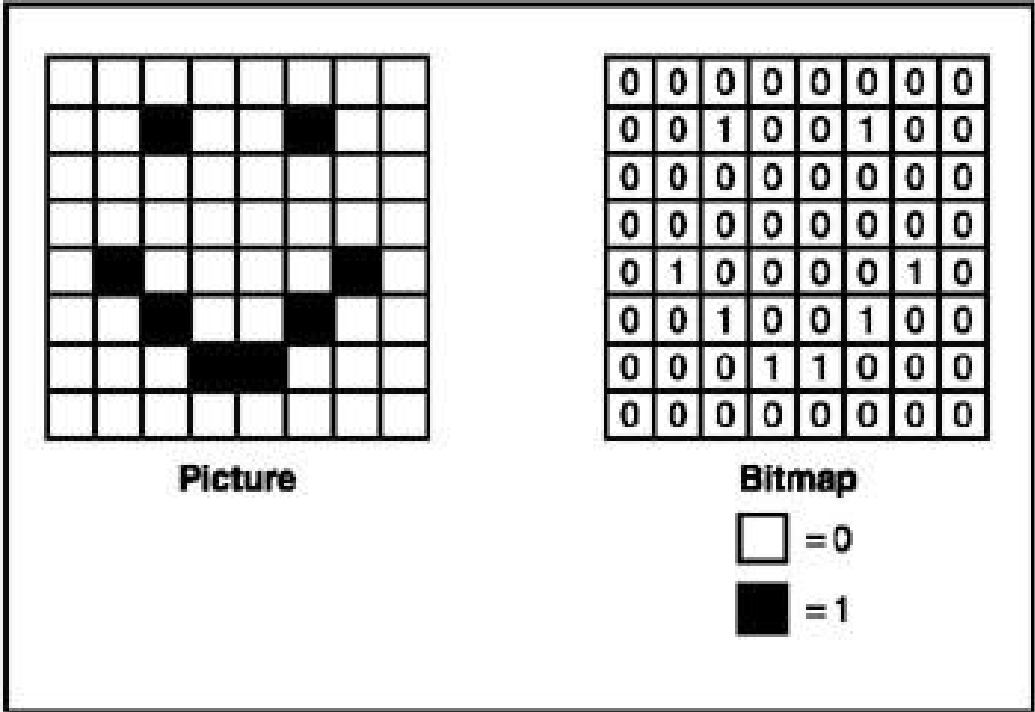
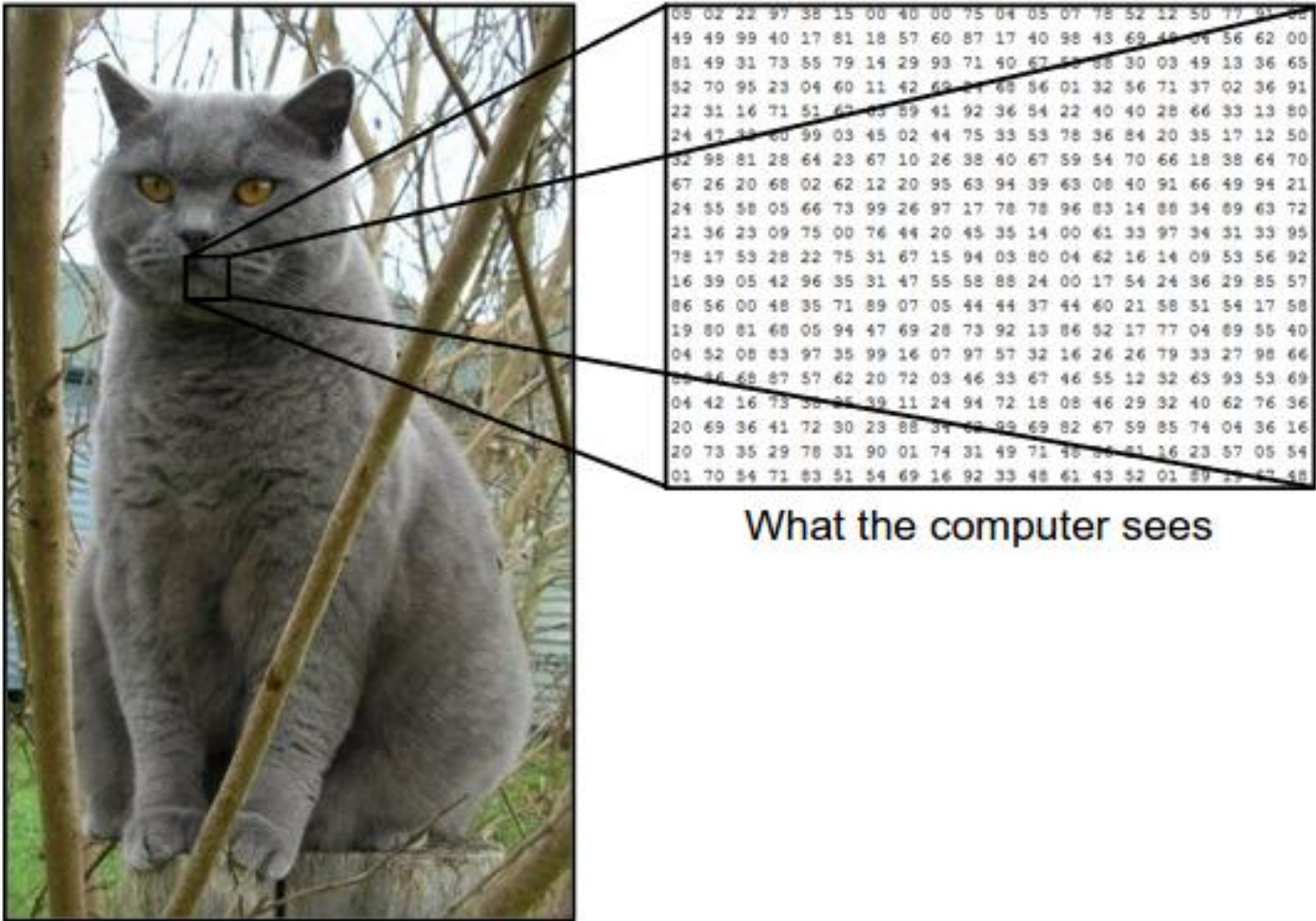


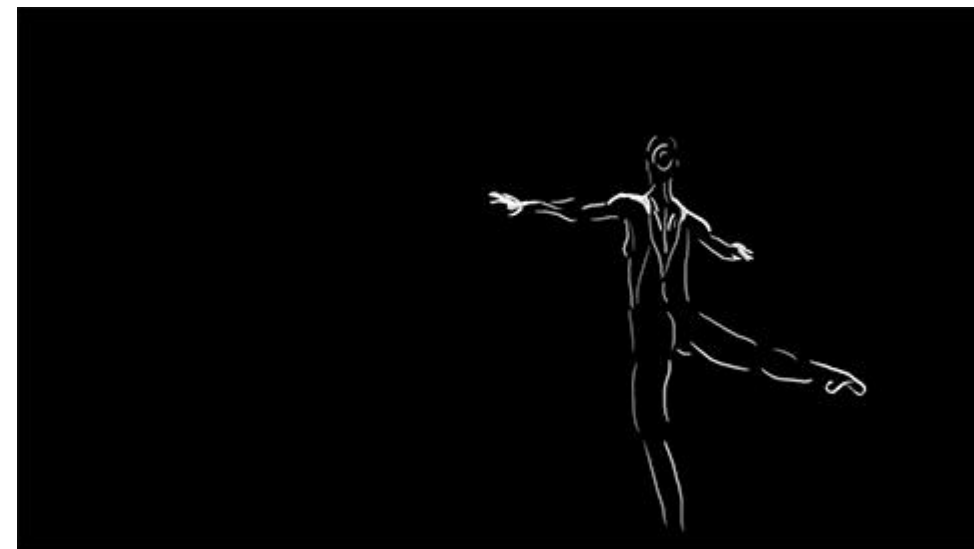
Image (Matrix)
画像 (行列)



Three-Dimensional (3D) Signal



Video (A collection of images)
ビデオ (画像のコレクション)

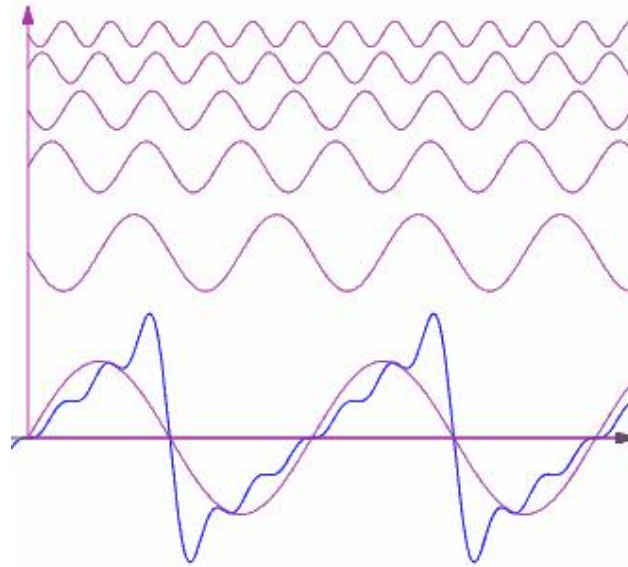


Animation (アニメーション)



1.1 Introduction of Fourier Analysis (フーリエ解析の紹介)

How to represent
complicated Signals?



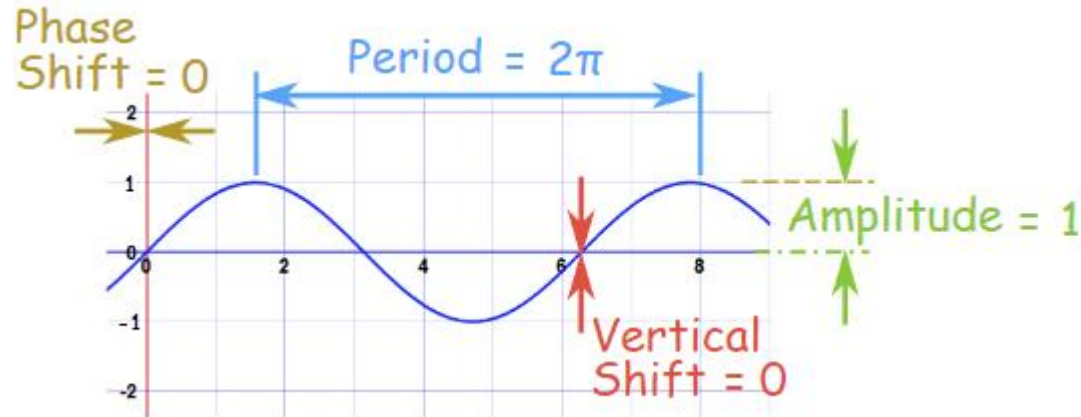
1.1 Introduction of Fourier Analysis (フーリエ解析の紹介)

Signal: $y = A \sin(Bx + C) + D$

- Amplitude (振幅) is A
- Period (周期) is $T = \frac{2\pi}{B}$
- phase shift (位相シフト) is $-\frac{C}{B}$
- vertical shift (垂直シフト) is D

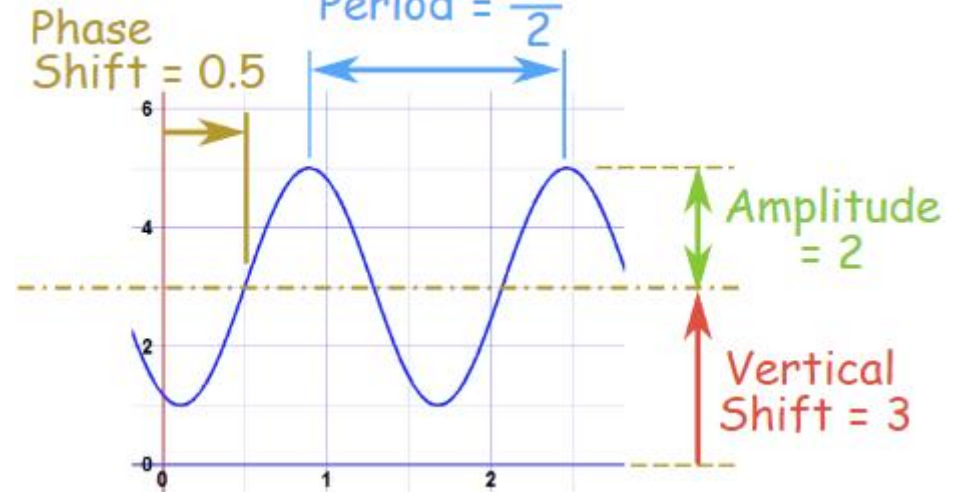
Example I: $\sin(x)$

- amplitude is $A = 1$
- period is $T = \frac{2\pi}{B} = 2\pi$
- phase shift is $-\frac{C}{B} = -\frac{0}{1} = 0$
- vertical shift is $D = 0$



Example II: $2 \sin(4x - 2) + 3$

- amplitude is $A = 2$
- period is $T = \frac{2\pi}{B} = \frac{2\pi}{4} = \frac{\pi}{2}$
- phase shift is $-\frac{C}{B} = -\frac{-2}{4} = \frac{1}{2} = 0.5$
- vertical shift is $D = 3$



1.1 Introduction of **Fourier Analysis** (フーリエ解析の紹介)

How to Represent Signals? 信号を表す方法は？

- **Joseph Fourier** had an amazing idea (1807):

Any periodic function can be rewritten as a weighted sum of **Sines** and **Cosines** of different frequencies.

- Don't believe it?
 - Neither did the Mathematicians Lagrange, Laplace, Poisson, etc.
 - Not translated into English until 1878!
- But it's true!
 - called **Fourier Series**
 - Possibly the greatest tool used in Engineering



Joseph Fourier (1768-1830)
ジョゼフ・フーリエ

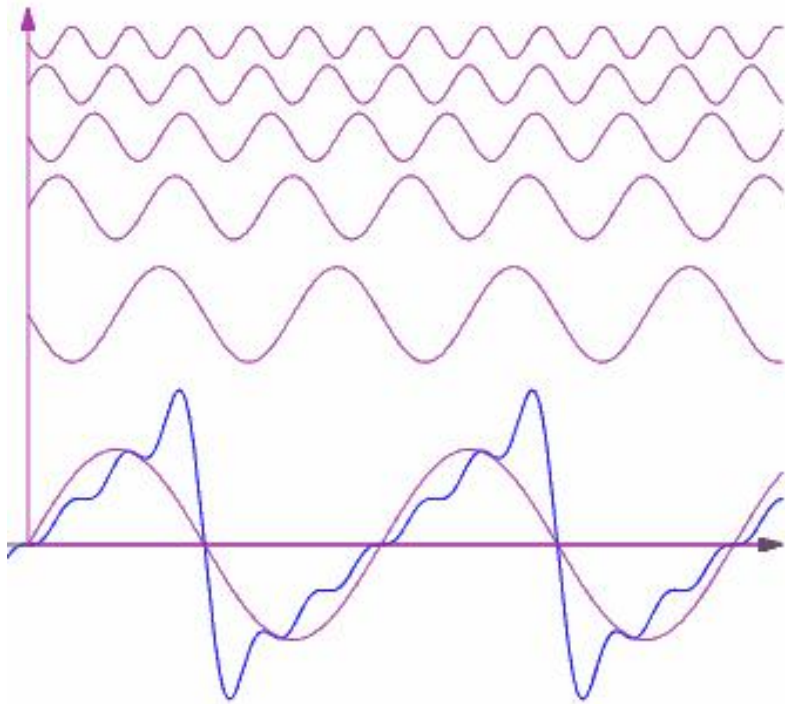
1.1 Introduction of Fourier Analysis (フーリエ解析の紹介)

How to represent complicated Signals?

Joseph Fourier concluded that an arbitrary signal can be represented as a sum of an infinite number of weighted sinusoids, i.e., sine and cosine signals.

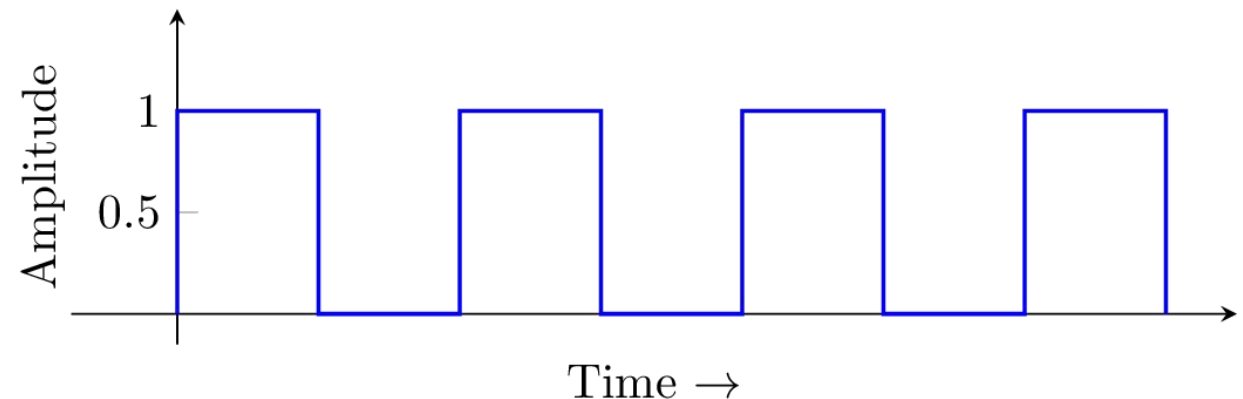


Joseph Fourier
(1768-1830)



Check the example for square wave.

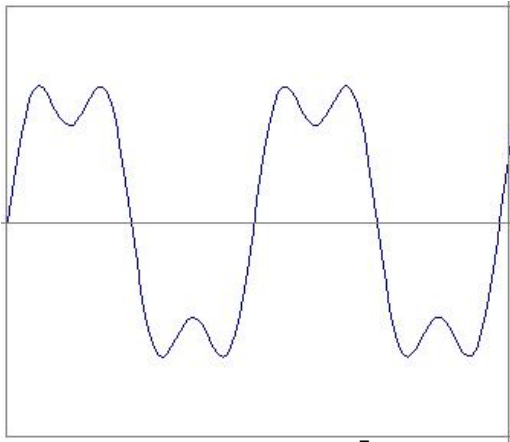
Square wave



Time and Frequency

- Example : $g(t) = \sin(2\pi f t) + \frac{1}{3}\sin(2\pi(3f)t)$

signal

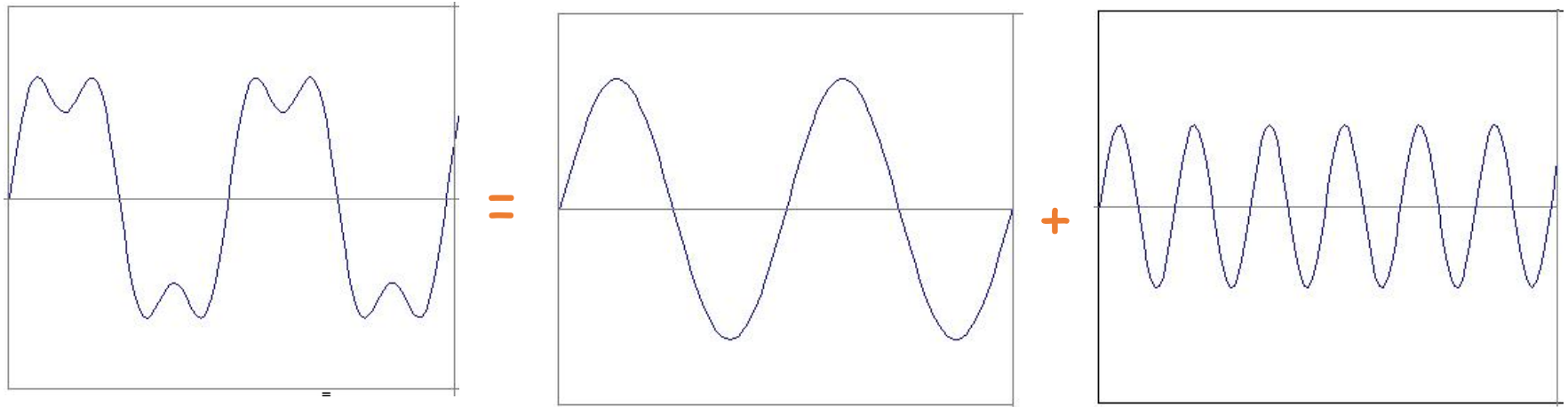


frequency Time

Time and Frequency

- Example : $g(t) = \sin(2\pi f t) + \frac{1}{3}\sin(2\pi(3f)t)$

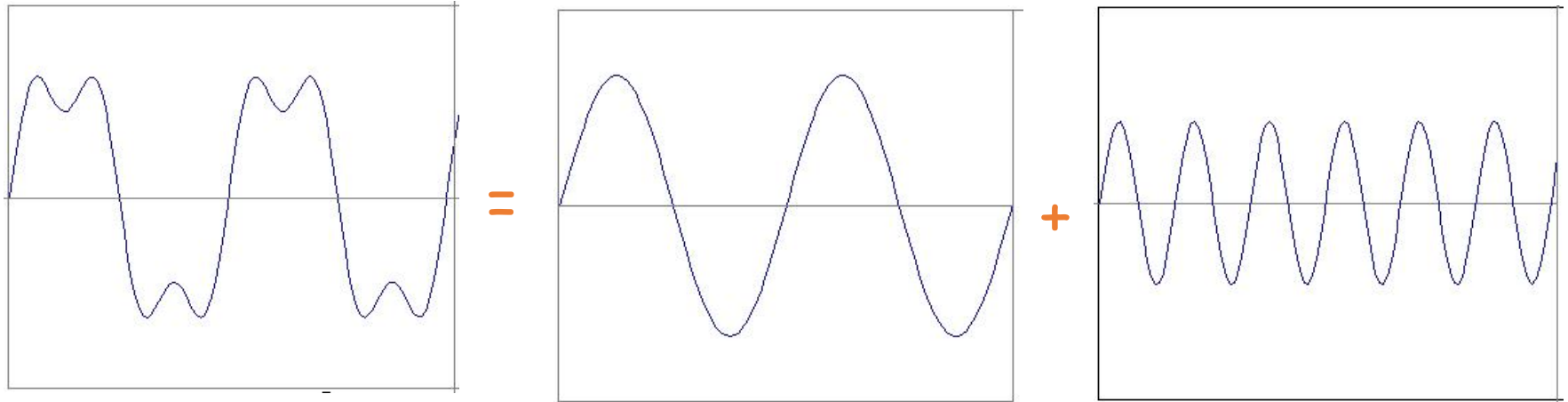
signal



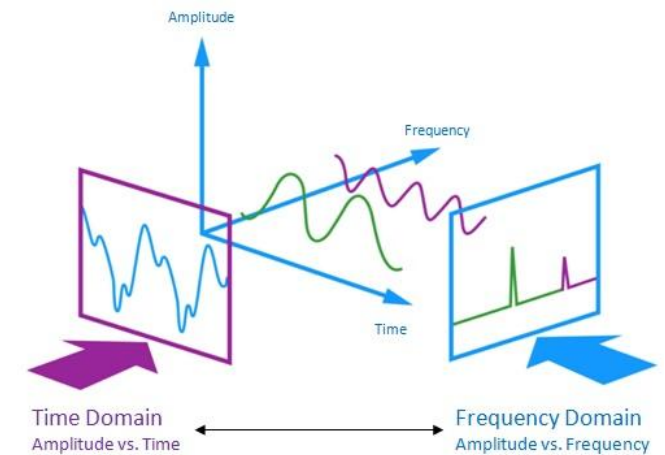
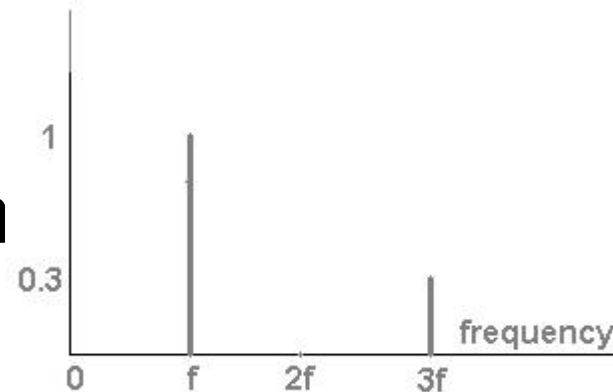
Time and Frequency

- Example : $g(t) = \sin(2\pi f t) + \frac{1}{3}\sin(2\pi(3f)t)$

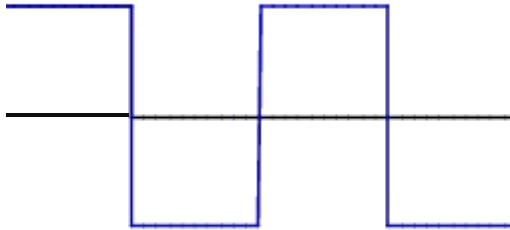
signal



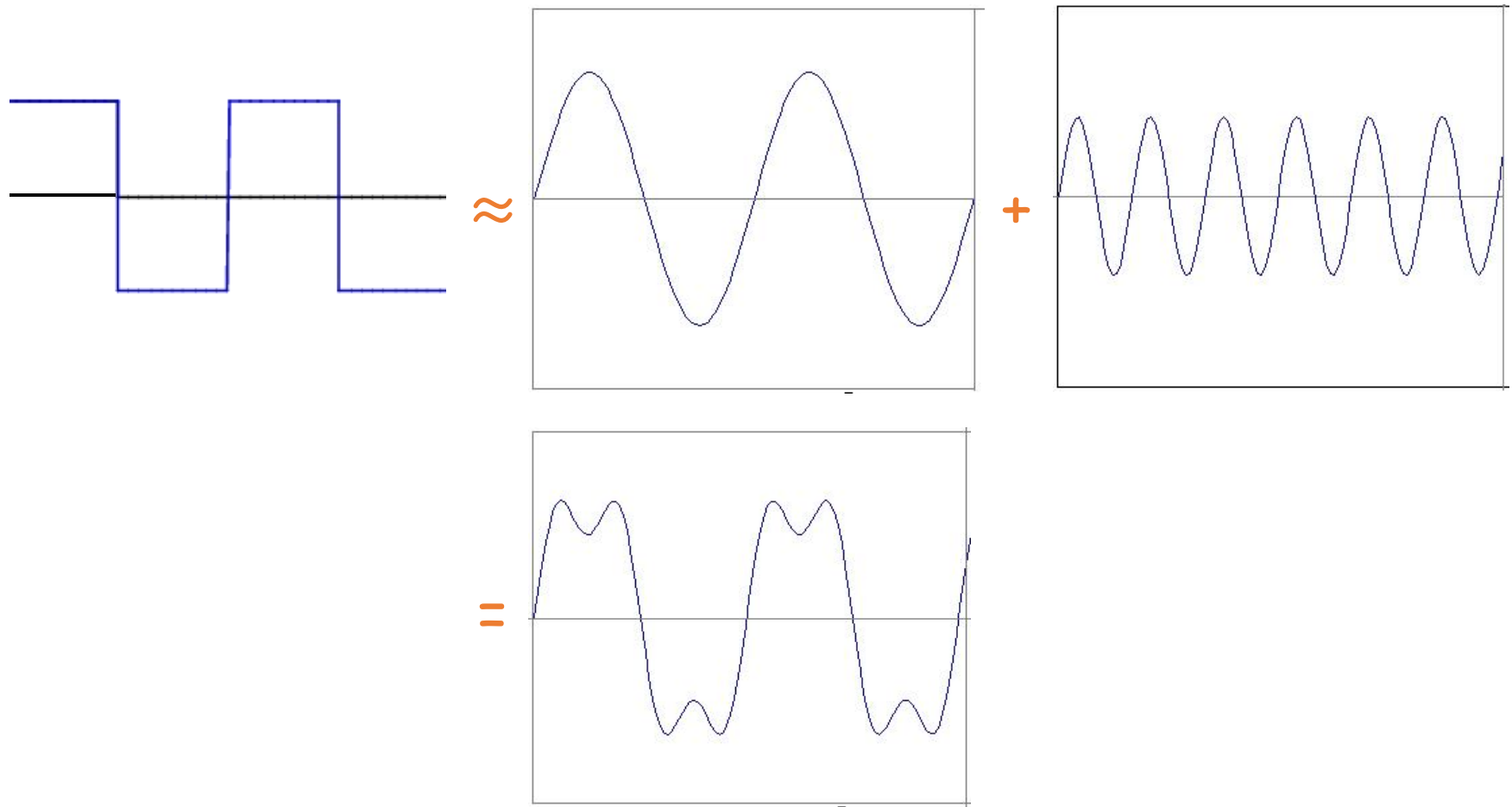
Frequency Spectrum



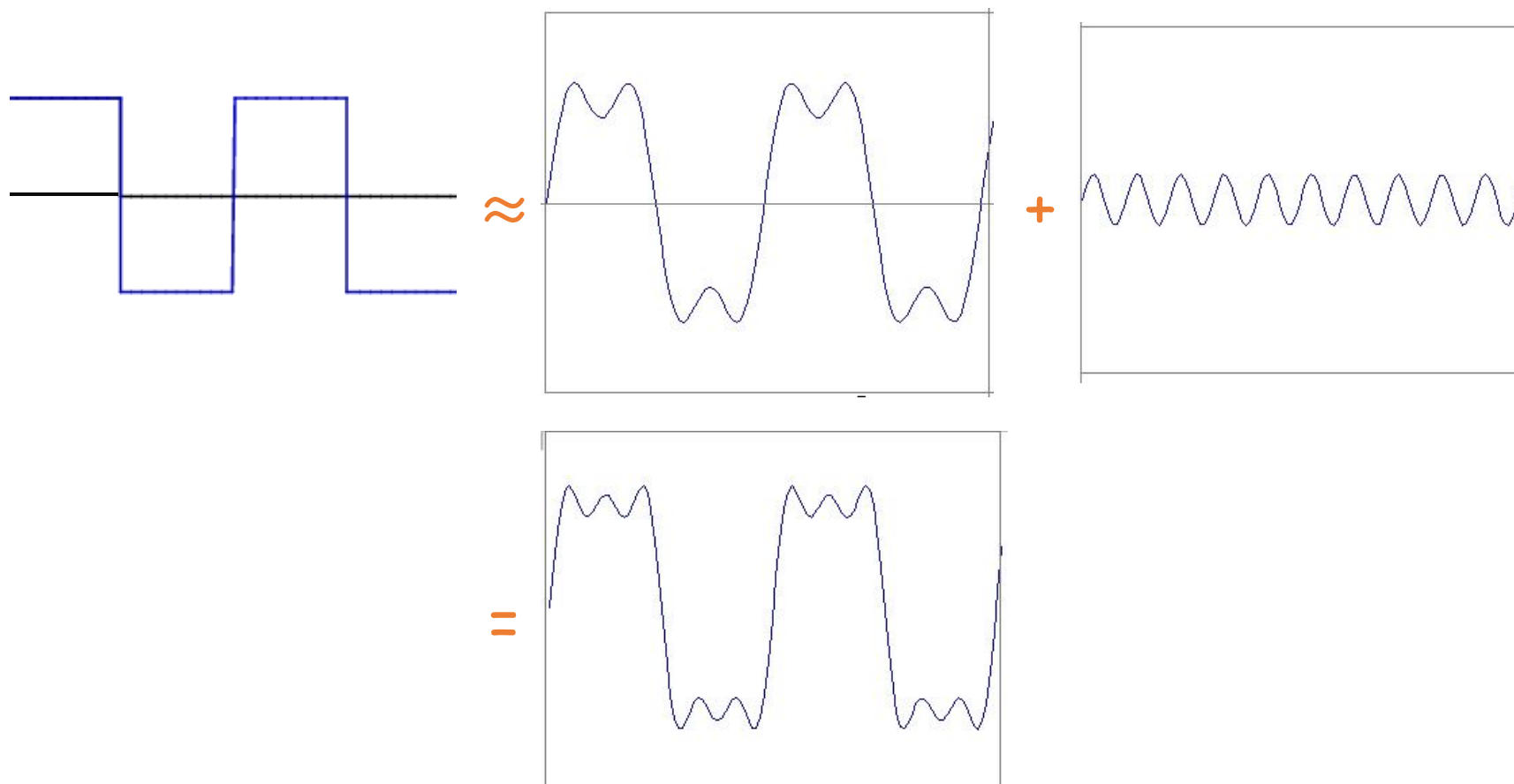
Time and Frequency



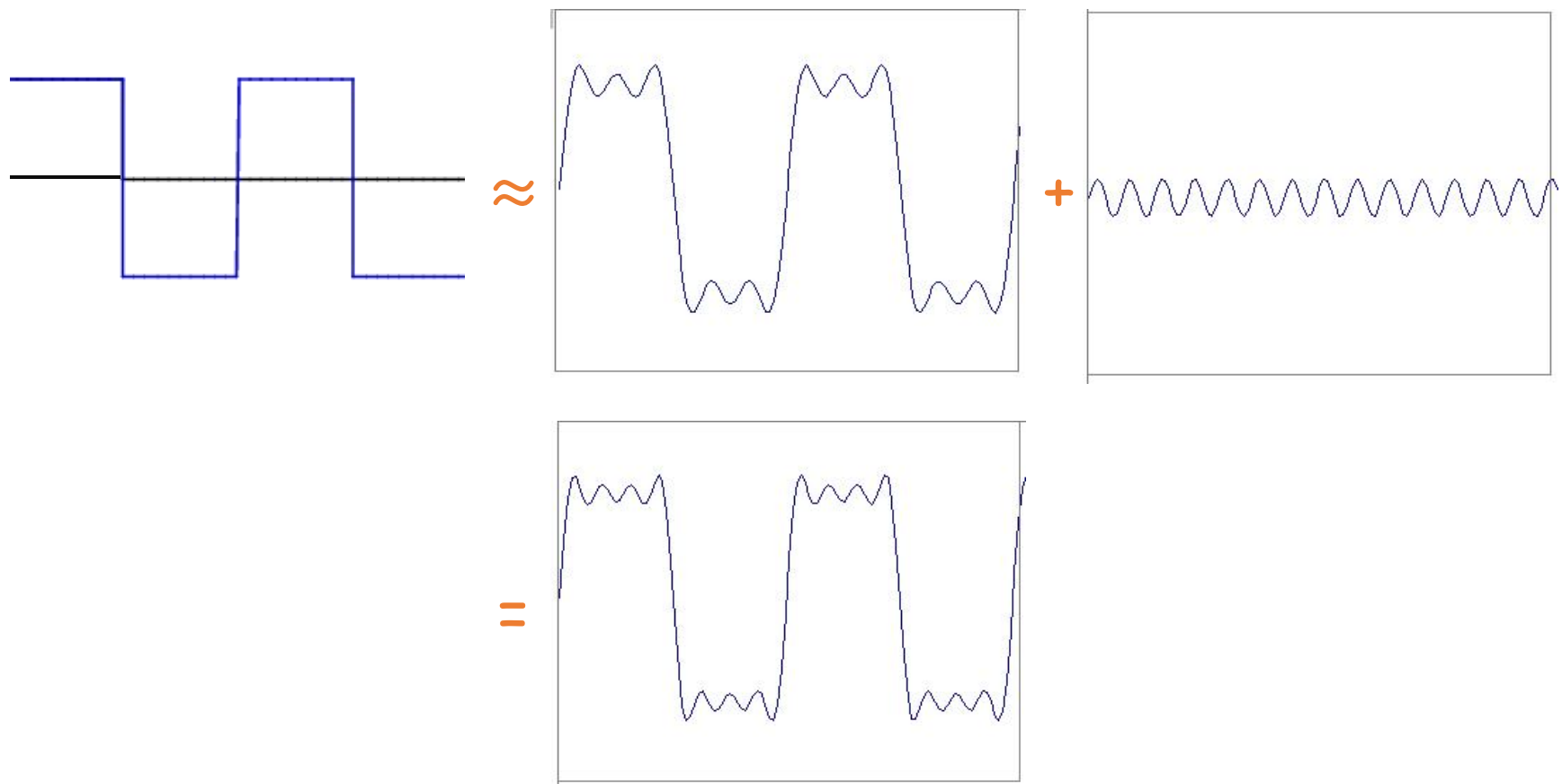
Time and Frequency



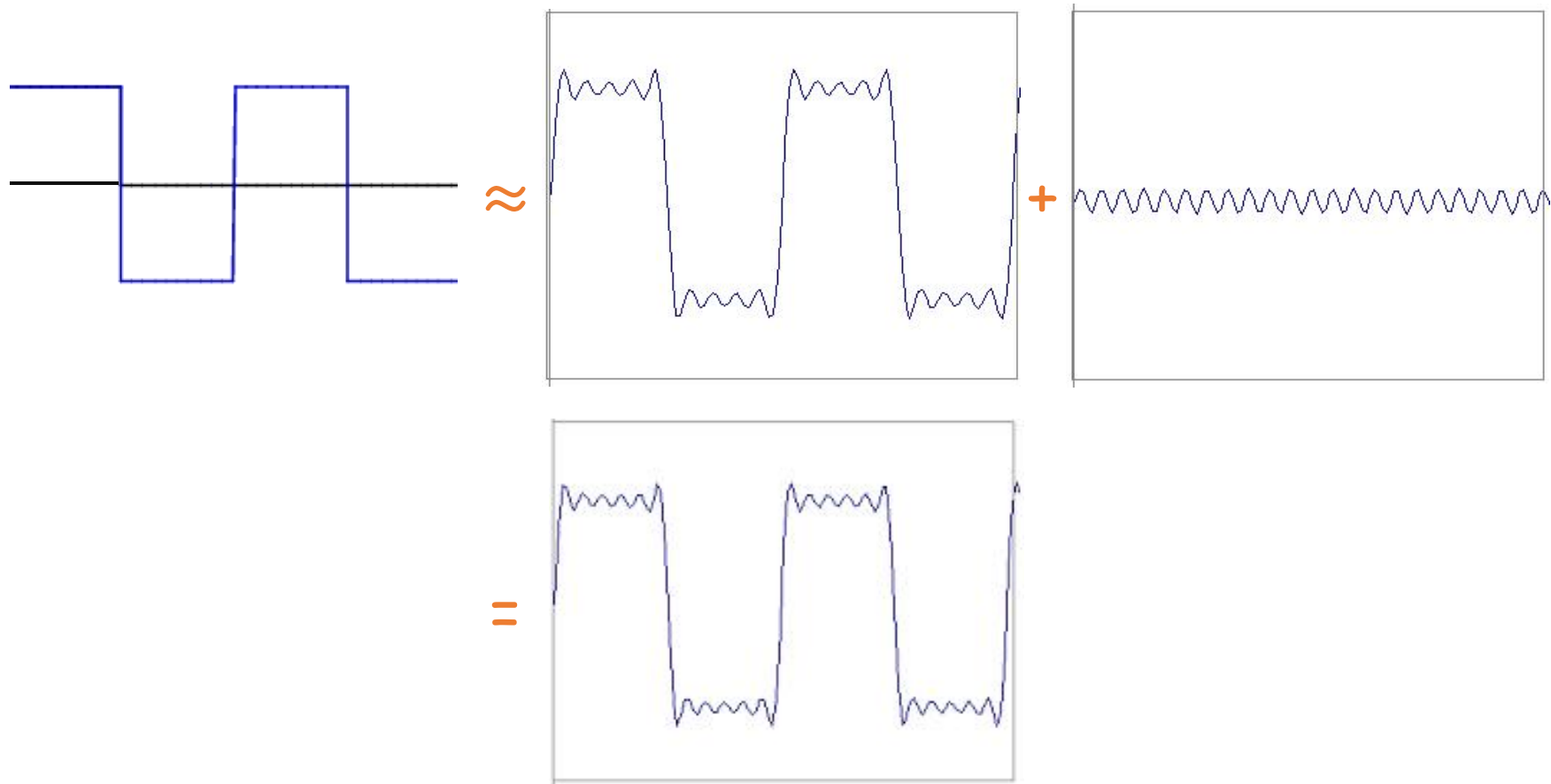
Time and Frequency



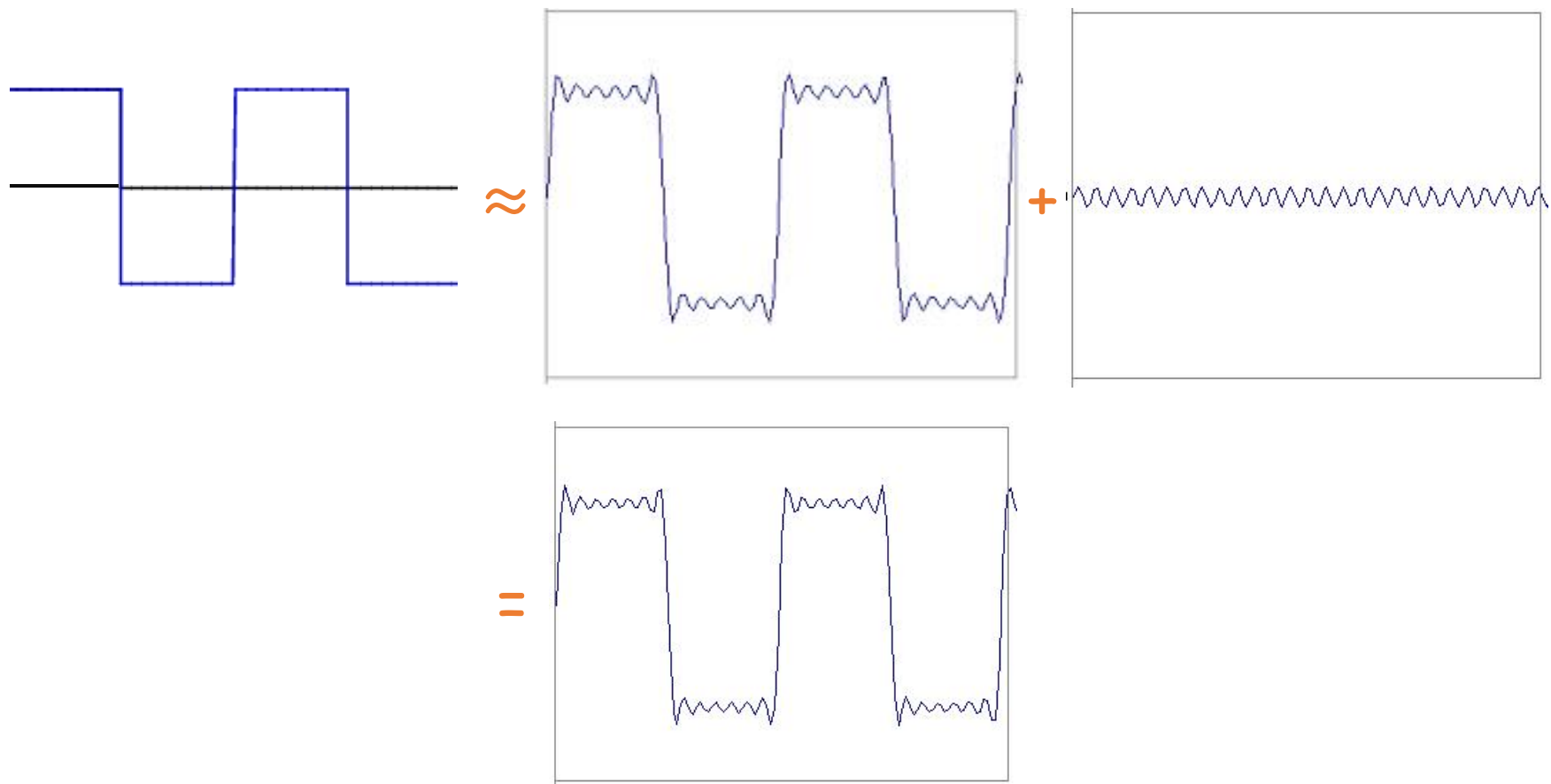
Time and Frequency



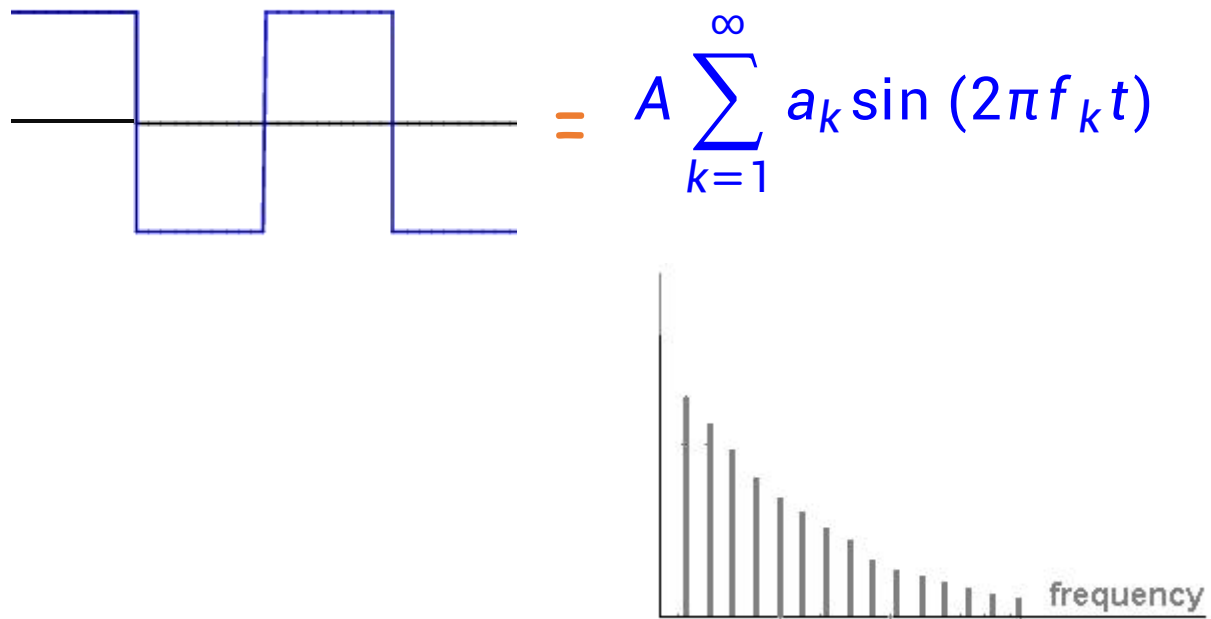
Time and Frequency



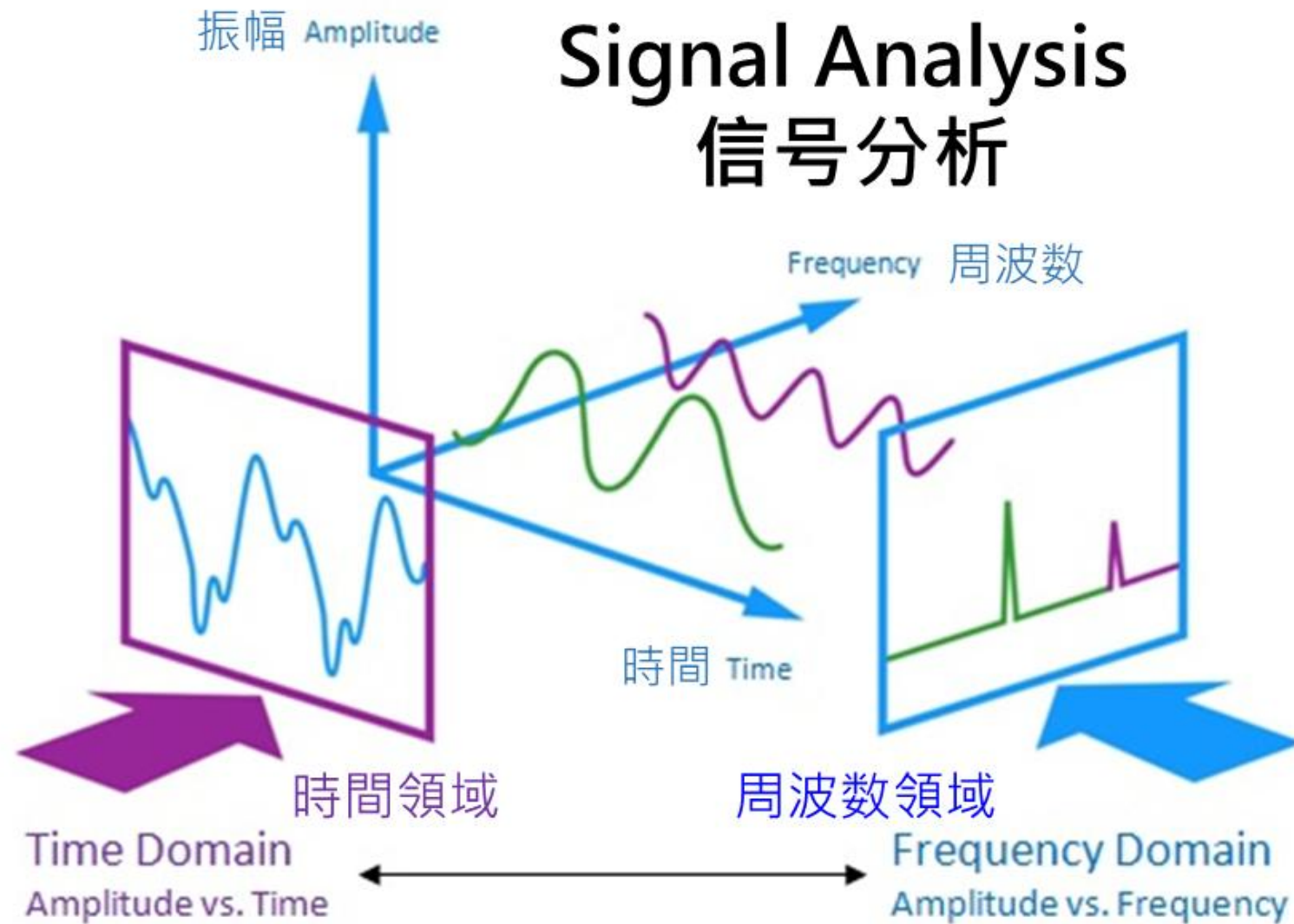
Time and Frequency



Time and Frequency

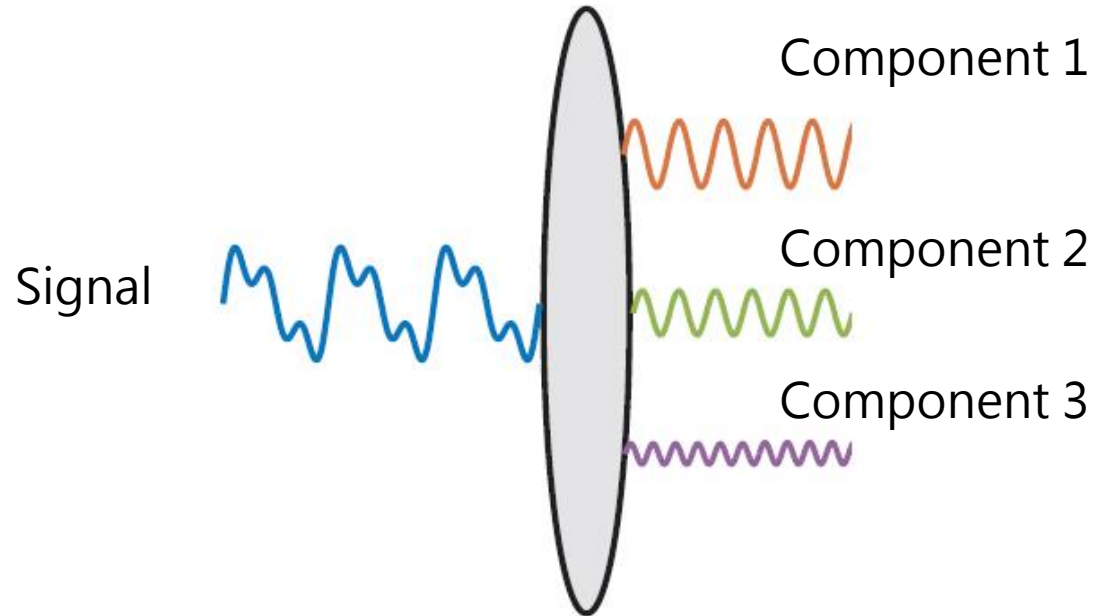


1.1 Introduction of Fourier Analysis (フーリエ解析の紹介)

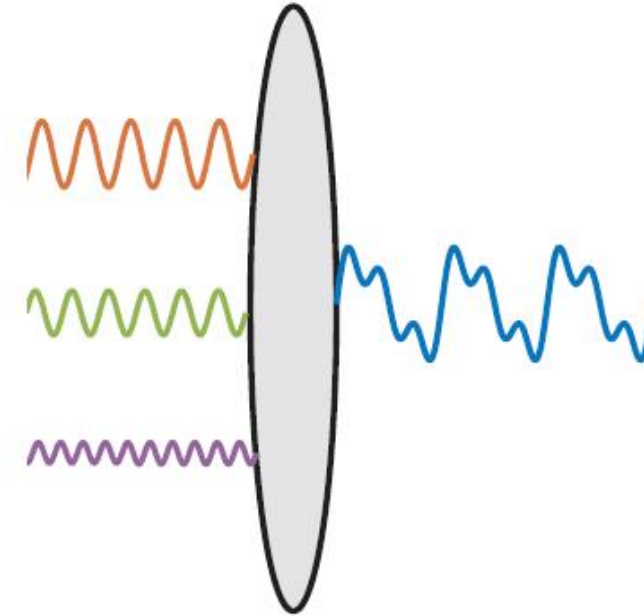


1.1 Introduction of Fourier Analysis (フーリエ解析の紹介)

Fourier Analysis



Fourier Synthesis



Fourier analysis is used to understand composite waves.

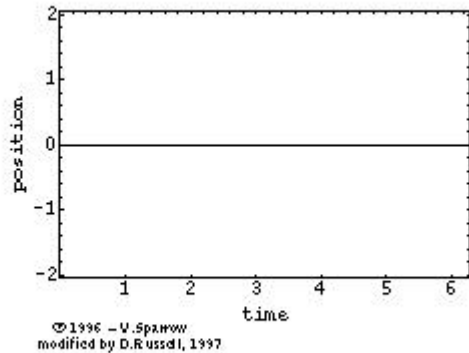
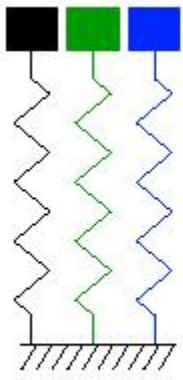
(a) Analysis: breaking a given signal into sine and cosine components

(b) Synthesis: adding certain sine and cosine to create a desired signal.

1.1 Introduction of Fourier Analysis (フーリエ解析の紹介)

Fourier series are usually the most suitable expansions of signals for solving certain classical problems in **applied mathematics**.

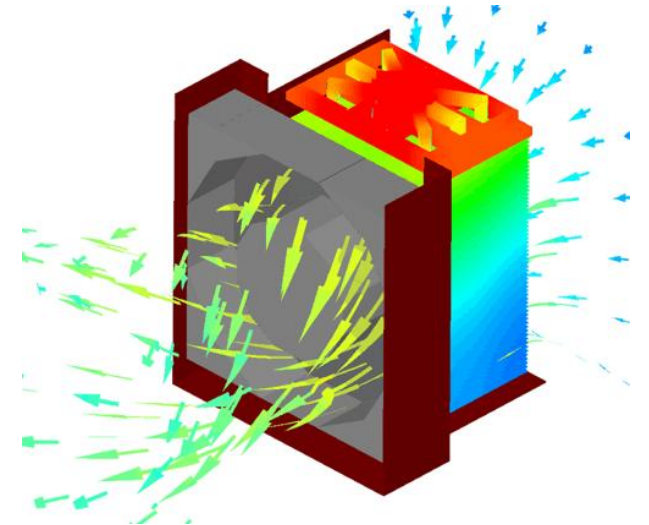
They are fundamental to the **important physical phenomena** (重要な物理現象), such as



mechanical vibrations



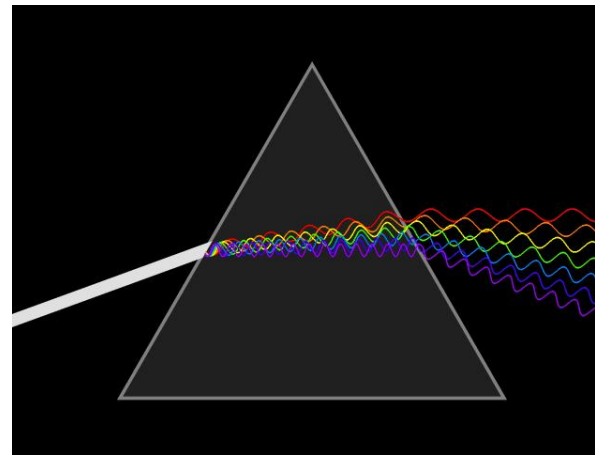
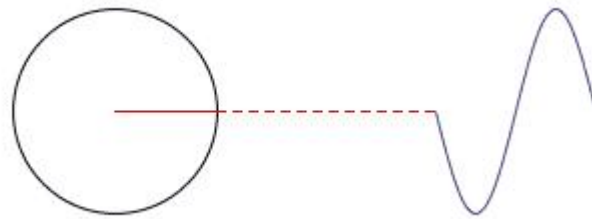
acoustic vibrations



heat transfer



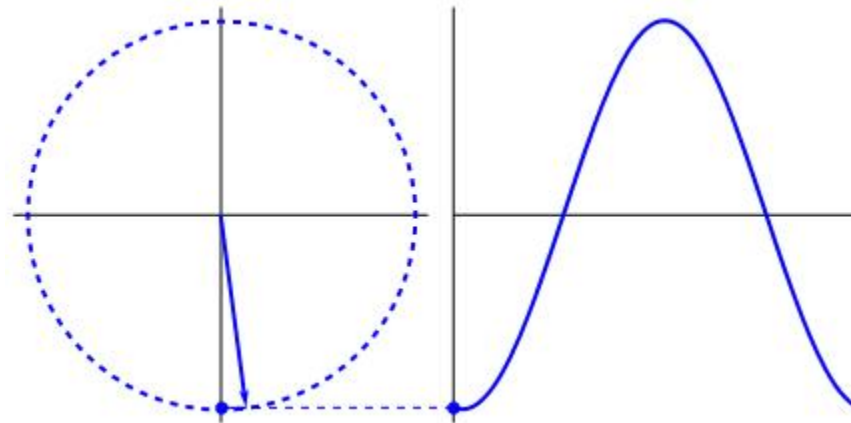
planetary motion



optics

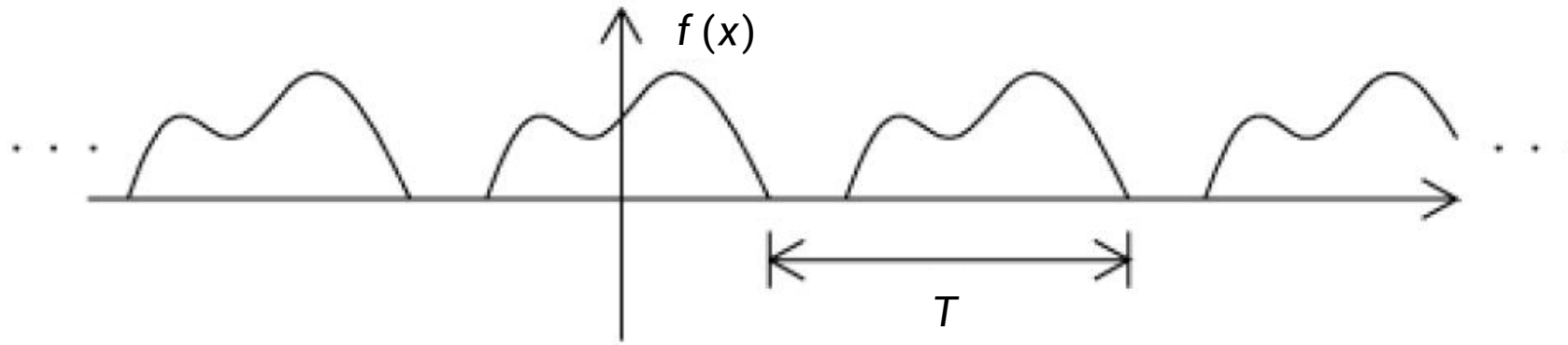
1.2 Periodic Functions

周期関数

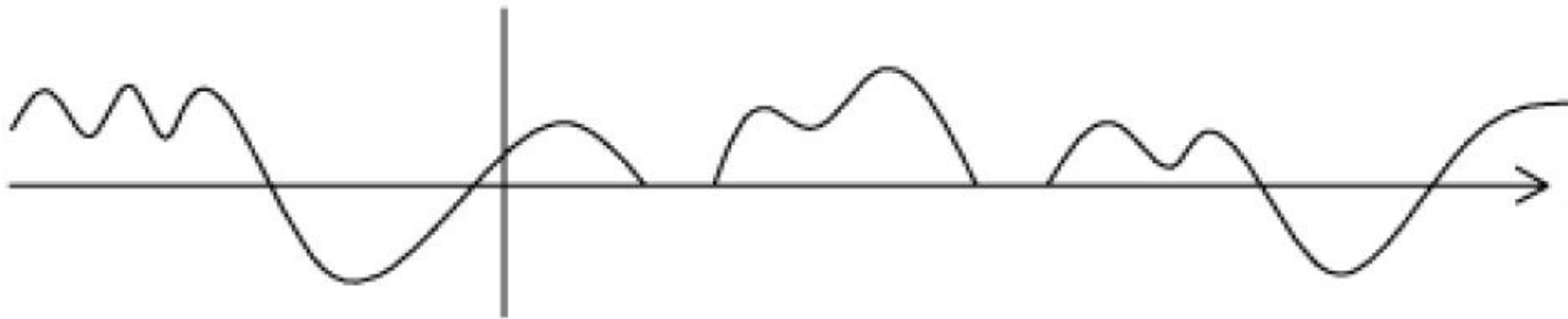


1.2 Periodic Functions (周期関数)

A **periodic** signal (周期信号) with period (周期) T



An **aperiodic** signal (非周期信号)



1.2 Periodic Functions (周期関数)

Periodic Functions

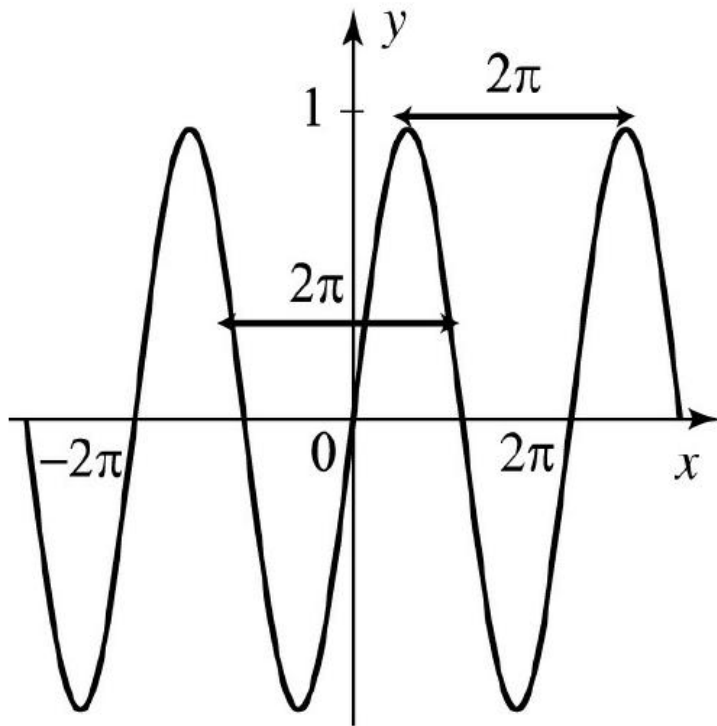


Figure 1.1 Graph of $\sin x$

This *periodicity* (周期性) is expressed by the identity (恒等)

$$\sin(x) = \sin(\underline{x + 2\pi}) \quad \text{for all } x$$

$\sin(x)$ repeat (繰り返す) every 2π units, its graph is obtained by repeating the portion over any interval of length 2π .

1.2 Periodic Functions (周期関数)

In general, a function f satisfying the identity

$$\sin(x) = \sin(x + T) \quad \text{for all } x \quad (1.1)$$

where $T > 0$, is called **periodic**, or more specifically, **T -periodic**.

The number T is called a **period of f** .

If f is nonconstant (非定数), we define the **fundamental period (基本周期)**, or **simply, the period (周期) of f** to be **the smallest positive number T** for which (1.1) holds.

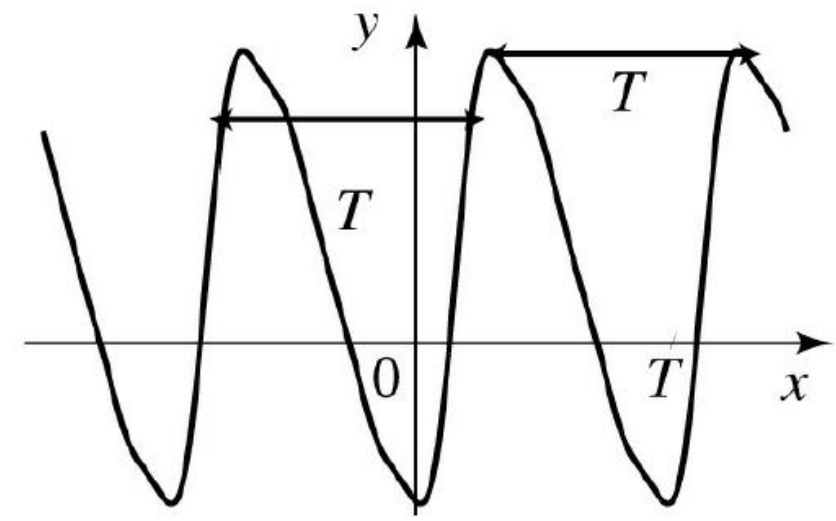


Figure 1.2 T -periodic function

1.2 Periodic Functions (周期関数)

$$\sin(x) = \sin(x + T) = \sin(x + 2T) = \dots = \sin(x + nT) = \dots$$

Hence if T is a period, then nT is also a period for any integer $n > 0$.

In the case of the sine function, this amounts to saying that $2\pi, 4\pi, 6\pi, \dots$ are all periods of $\sin(x)$, but only 2π is the fundamental period.

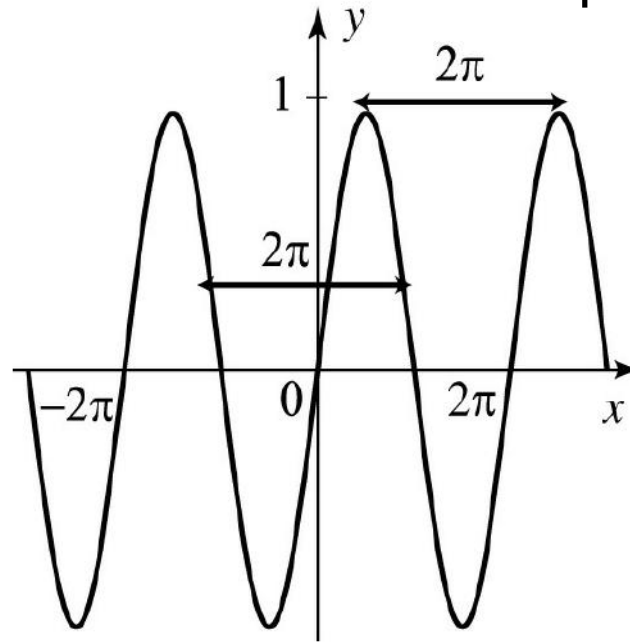


Figure 1.1 Graph of $\sin x$
MA05 Fourier Analysis

1.2 Periodic Functions (周期関数)

Example 1.1 Describing a periodic function

Describe the 2-periodic function f in Figure 1.3 in two different ways:

- (a) By considering its values on the interval (間隔) $0 \leq x < 2$
- (b) By considering its values on the interval (間隔) $-1 \leq x < 1$

$$T = 2$$

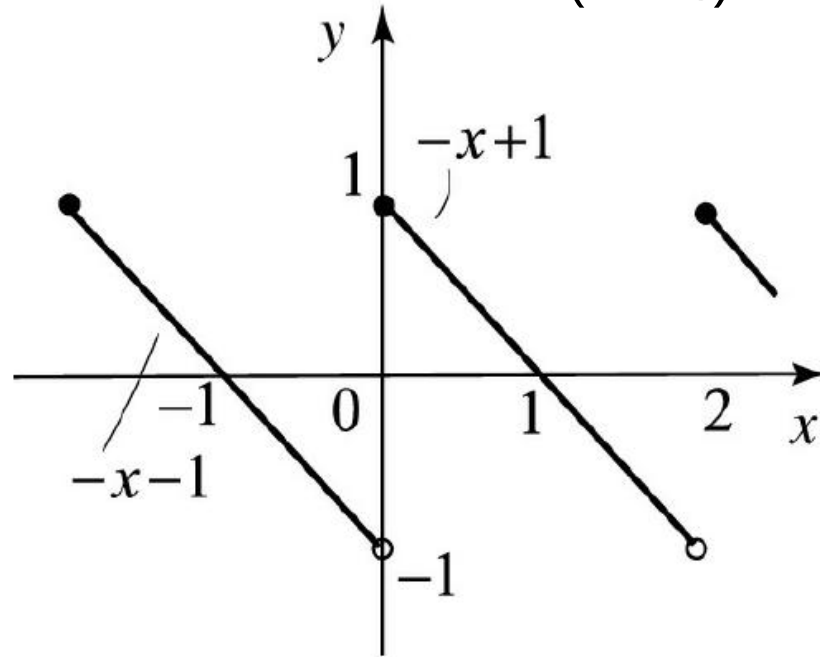


Figure 1.3 A 2-periodic function

1.2 Periodic Functions (周期関数)

Solution:

(a) On the interval $0 \leq x < 2$ the graph is a portion of the straight line $y = -x + 1$.

Thus $f(x) = -x + 1$ if $0 \leq x < 2$

(b) On the interval $-1 \leq x < 1$, the graph consists of two straight lines (Figure 1.3). We have

$$f(x) = \begin{cases} -x - 1 & \text{if } -1 \leq x < 0 \\ -x + 1 & \text{if } 0 \leq x < 1 \end{cases}$$

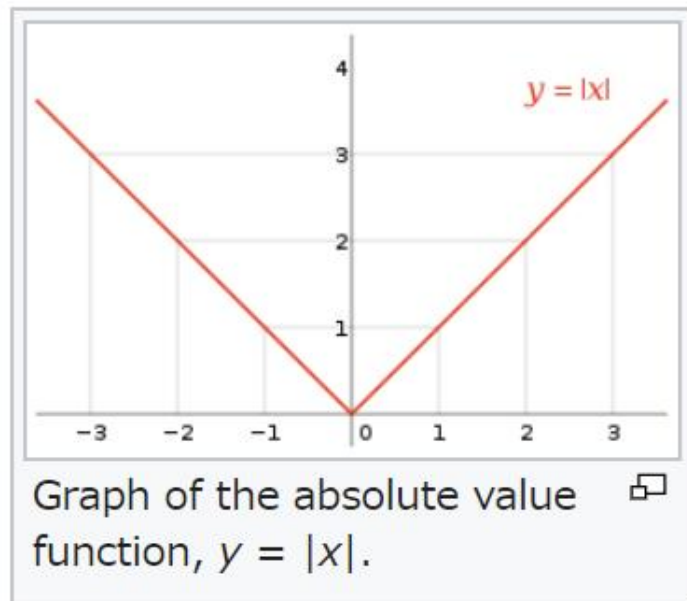
As in part (a), the relation $f(x + 2) = f(x)$ describes f for all values of x outside the interval $[-1, 1)$.

Although the formulas in Example 1.1(a) and (b) are different, they describe the same periodic function.

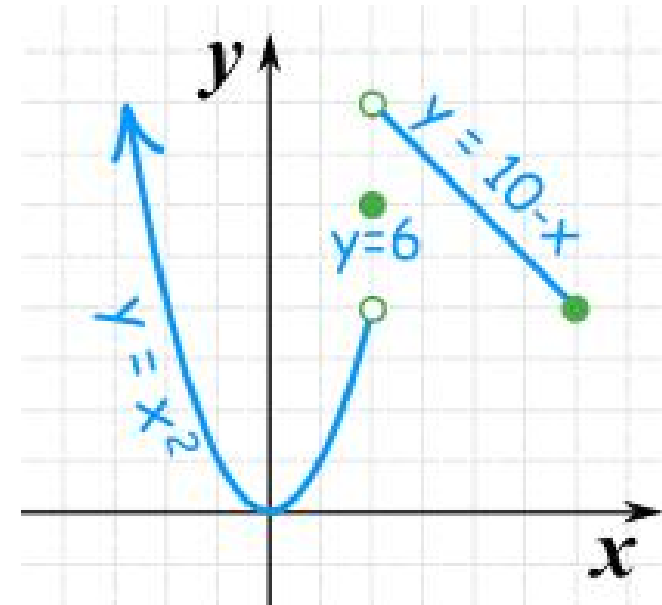
1.2 Periodic Functions (周期関数)

A **piecewise-defined function** is defined by multiple sub-functions, each sub-function applying to a certain interval of the main function's domain, a sub-domain

$$|x| = \begin{cases} -x, & \text{if } x < 0 \\ x, & \text{if } x \geq 0 \end{cases}$$



$$f(x) = \begin{cases} x^2 & \text{if } x < 2 \\ 6 & \text{if } x = 2 \\ 10 - x & \text{if } x > 2 \text{ and } x \leq 6 \end{cases}$$



1.2 Periodic Functions (周期関数)

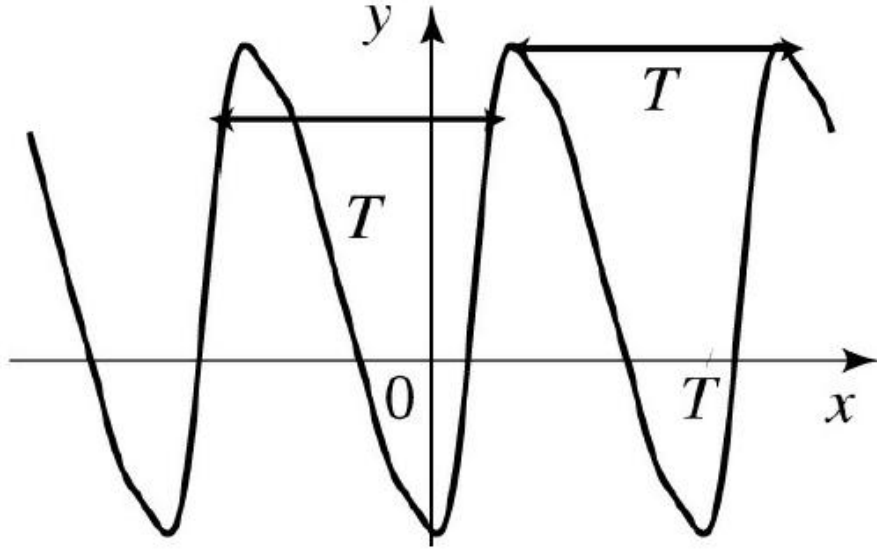


Figure 1.2 T -periodic function

continuous (連続)

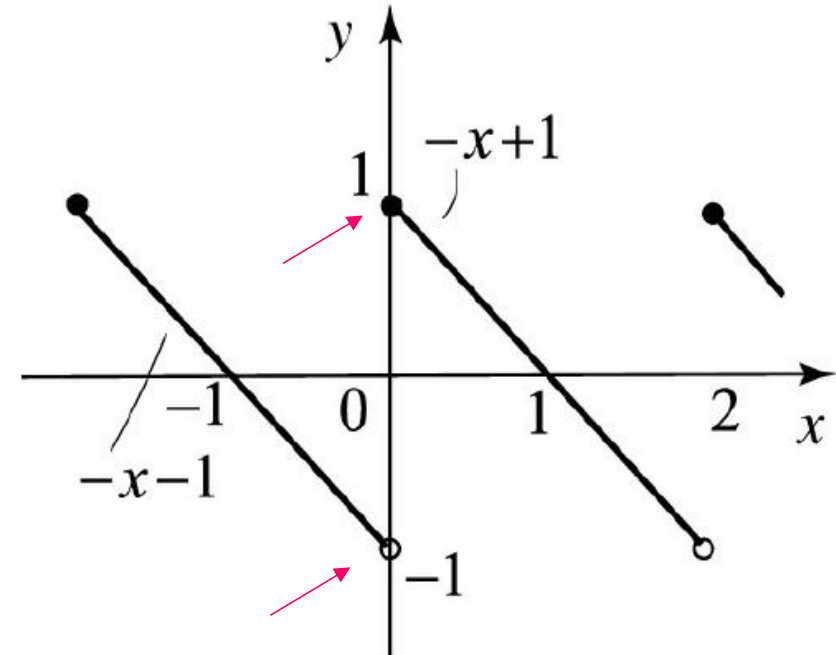


Figure 1.3 A 2-periodic function

discontinuous (不連続)

1.2 Periodic Functions (周期関数)

Discontinuity (不連続性)

Consider the function $f(x)$ in Figure 1.3. This function is not continuous at $x = 0, \pm 2, \pm 4, \dots$. Take a point of discontinuity, say $x = 0$. The limit of the function from the left is -1 , while the limit from the right is 1 .

$$f(0-) = \lim_{x \rightarrow 0^-} f(x) = -1 \text{ and } f(0+) = \lim_{x \rightarrow 0^+} f(x) = 1$$

In general, we write

$$f(c-) = \lim_{x \rightarrow c^-} f(x)$$

(1.2)

$$f(c+) = \lim_{x \rightarrow c^+} f(x)$$

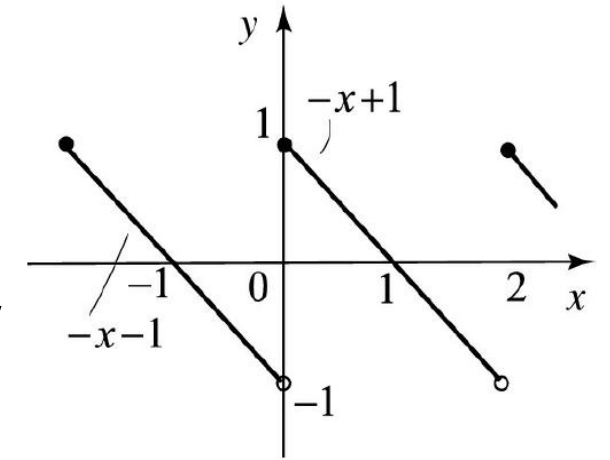


Figure 1.3 A 2-periodic function

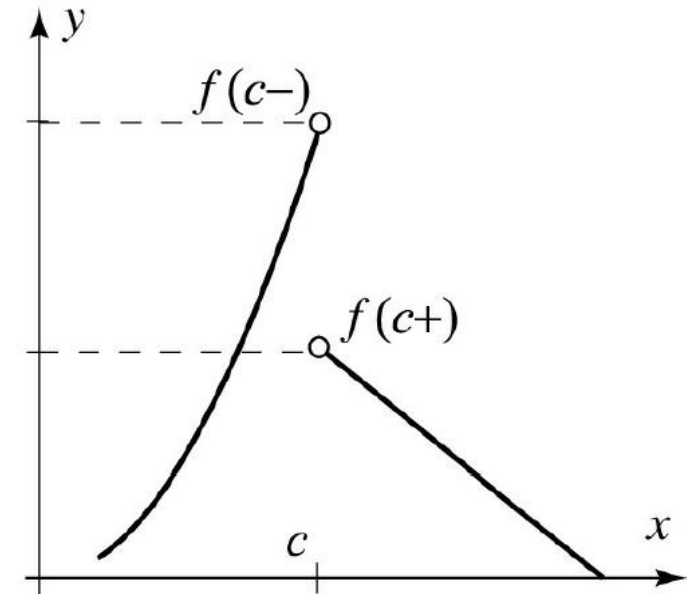


Figure 1.4 Left and right limits.42

1.2 Periodic Functions (周期関数)

Piecewise **Continuous** Functions (区分的連続関数)

A function f is said to be piecewise continuous on the interval $[a, b]$ if $f(a+)$ and $f(b-)$ exist, and f is defined and continuous on (a, b) except at a finite number of points in (a, b) where the left and right limits exist.

A periodic function is said to be **continuous** if it is continuous on the entire real line.

1.2 Periodic Functions (周期関数)

Piecewise Continuous Functions (区分的連続関数)

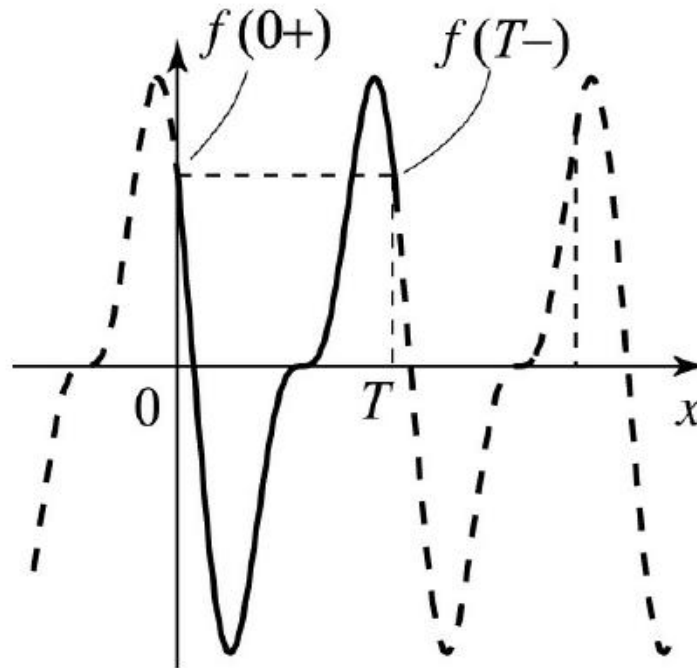


Figure 1.5 A continuous T -periodic function

At endpoints of the periodic function, if f is T -periodic and continuous, then necessarily $f(0+) = f(T-)$.

1.2 Periodic Functions (周期関数)

Piecewise Smooth Functions (区分的平滑関数)

A function f , defined on the interval $[a, b]$, is said to be **piecewise smooth** if f and f' are **piecewise continuous** on $[a, b]$. (Here f' is derivative (導関数) of f .)

Thus f is **piecewise smooth** if f is piecewise continuous on $[a, b]$,

f' exists and is continuous in (a, b) except possibly at finitely many points c

where the one-sided limits $\lim_{x \rightarrow c^-} f'(x)$ and $\lim_{x \rightarrow c^+} f'(x)$ exist.

Furthermore, $\lim_{x \rightarrow a^+} f'(x)$ and $\lim_{x \rightarrow b^-} f'(x)$ exist.

A function f is **smooth** if f and f' are continuous.

One-Variable Calculus – Definite integral of f

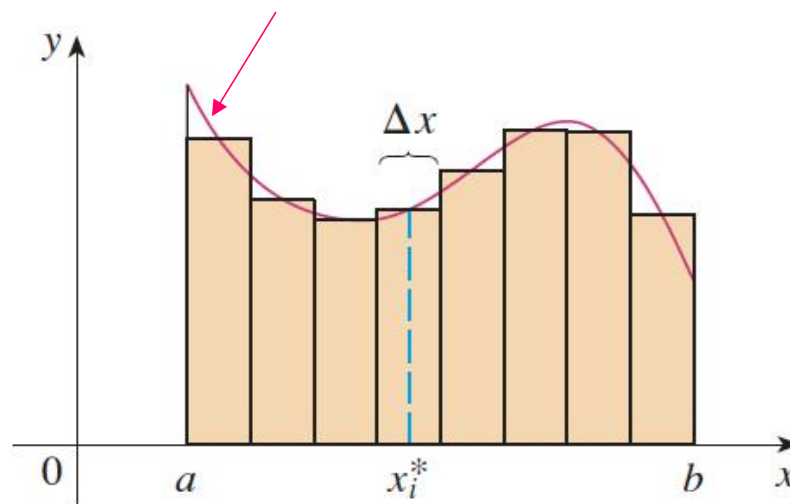
Riemann Sum

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x = \lim_{n \rightarrow \infty} [f(x_1^*) \Delta x + f(x_2^*) \Delta x + \dots + f(x_n^*) \Delta x]$$

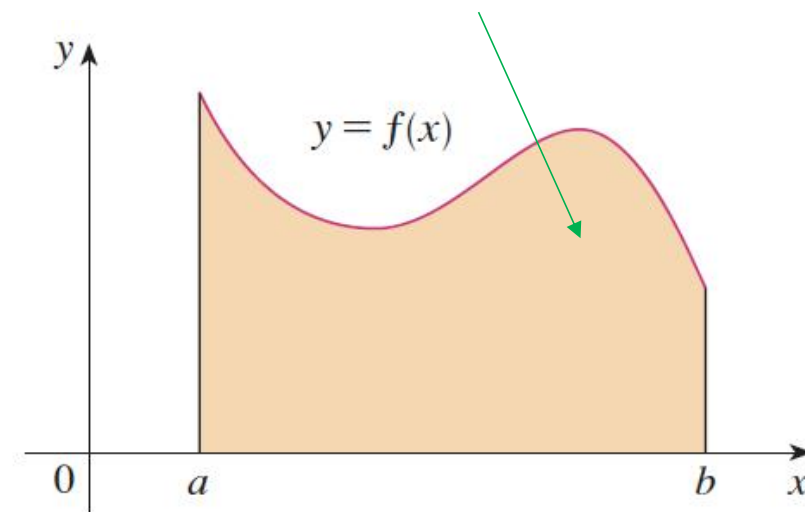
One-Variable Function

$$y = f(x)$$

$$y = f(x)$$



$$Area = \int_a^b f(x) dx$$



1.2 Periodic Functions (周期関数)

Theorem (定理) 1.1: Integral (積分) over one period

Suppose that f is piecewise continuous and T -periodic. Then, for any real number a , we have

$$\int_0^T f(x) dx = \int_a^{a+T} f(x) dx \quad (1.3)$$

1.2 Periodic Functions (周期関数)

Proof:

$$\begin{aligned} \text{(a)} \quad \int_{nT}^{(n+1)T} f(x) dx &= \int_0^T f(s + nT) ds \quad (\text{let } x = s + nT, dx = ds) \\ &= \int_0^T f(s) ds \quad (\text{because } f \text{ is } T\text{-periodic}) \\ &= \int_0^T f(x) dx \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad \int_{(n+1)T}^{a+T} f(x) dx &= \int_{nT}^a f(s + T) ds \quad (\text{let } x = s + T, dx = ds) \\ &= \int_{nT}^a f(s) ds \quad (\text{because } f \text{ is } T\text{-periodic}) \\ &= \int_{nT}^a f(x) dx \end{aligned}$$

$$\begin{aligned} \text{(c)} \quad \int_a^{a+T} f(x) dx &= \int_a^{(n+1)T} f(x) dx + \int_{(n+1)T}^{a+T} f(x) dx \\ &= \int_a^{(n+1)T} f(x) dx + \int_{nT}^a f(x) dx \quad (\text{by (b)}) \\ &= \int_{nT}^{(n+1)T} f(x) dx = \int_0^T f(x) dx \quad (\text{by (a)}) \end{aligned}$$



1.2 Periodic Functions (周期関数)

Example 1.2 Integrating periodic functions

Let f be the 2-periodic function in Example 1.1. Use Theorem 1.1 to compute

$$\int_{-1}^1 f^2(x) dx$$

Solution:

Observe that $f^2(x)$ is also 2-periodic.

Thus, by Theorem 1.1, to compute this integral we may choose interval of length 2.

Since on the interval $(0,2)$ the function $f(x)$ is given by a single formula, we choose to work on this interval, and, using the formula from Example 1.1, we find

$$\int_{-1}^1 f^2(x) dx = \int_0^2 f^2(x) dx = \int_0^2 (-x+1)^2 dx = -\frac{1}{3}(-x+1)^3 \Big|_0^2 = \frac{2}{3}$$

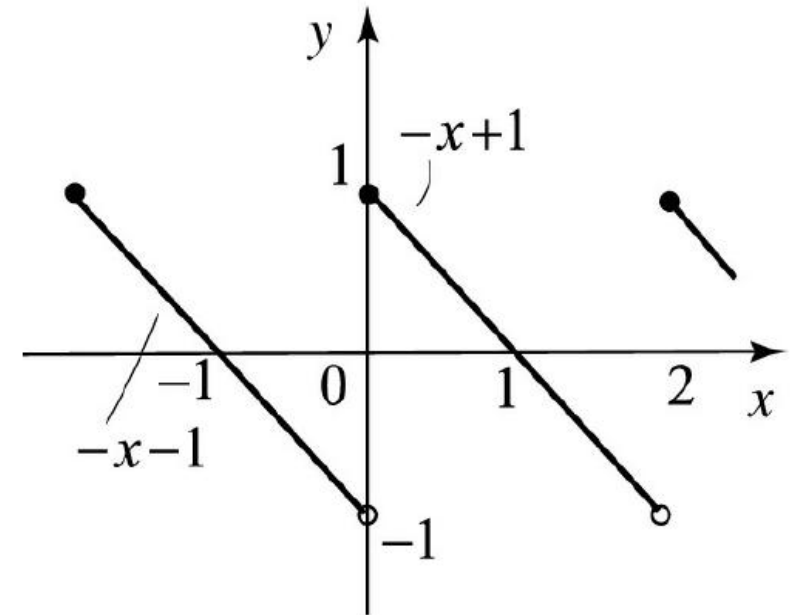


Figure 1.3 A 2-periodic function

1.2 Periodic Functions (周期関数)

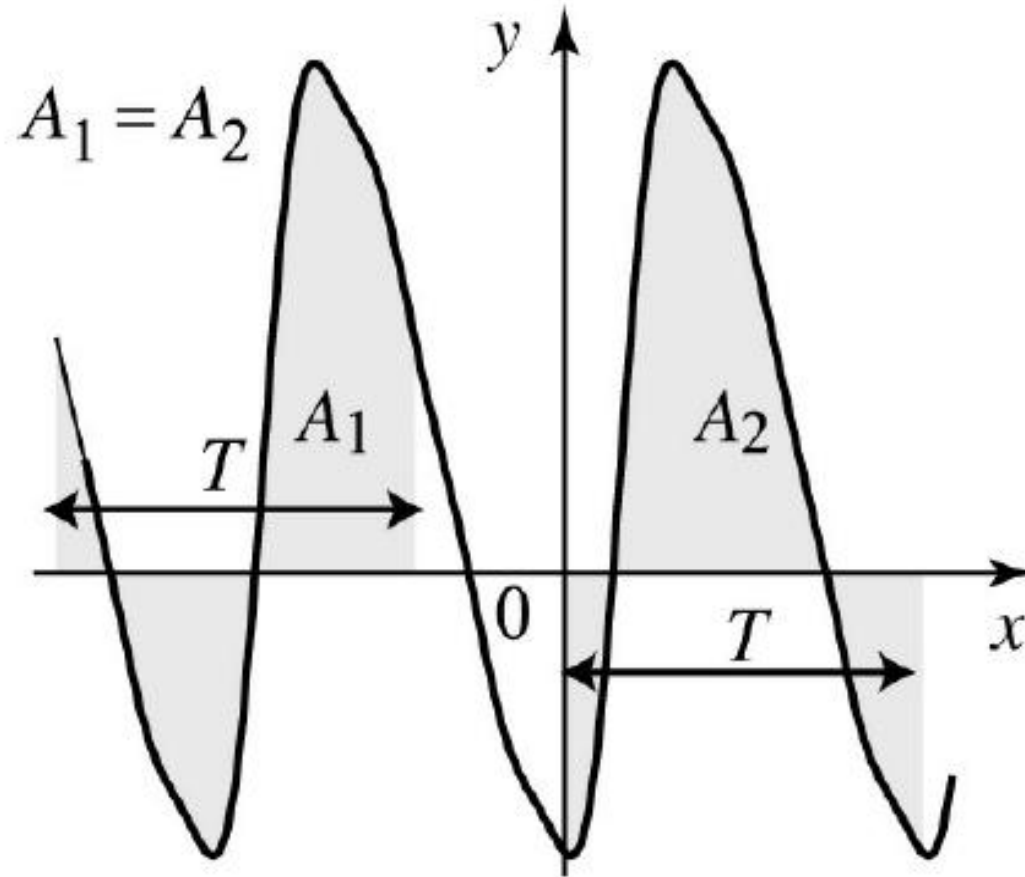


Figure 1.6 Areas (面積) over one period.

1.3 The Trigonometric System and Orthogonality

三角法システムと直交性

1.3 The Trigonometric System and Orthogonality (三角法システムと直交性)

The most important periodic functions are those in the $(2\pi$ -periodic) **trigonometric system**

$$1, \cos x, \cos 2x, \cos 3x, \dots, \cos mx, \dots, \\ \sin x, \sin 2x, \sin 3x, \dots, \sin nx, \dots$$

Useful property: **orthogonality**

We say that **two functions $f(x)$ and $g(x)$ are orthogonal over the interval $[a, b]$** if

$$\int_a^b f(x)g(x) dx = 0$$

1.3 The Trigonometric System and Orthogonality (三角法システムと直交性)

Dot Product (ドット積)

Scalar product

Definition: If $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$ and $\mathbf{b} = \langle b_1, b_2, b_3 \rangle$, then the dot product of $\mathbf{a}$ and $\mathbf{b}$ is the number $\mathbf{a} \cdot \mathbf{b}$ given by

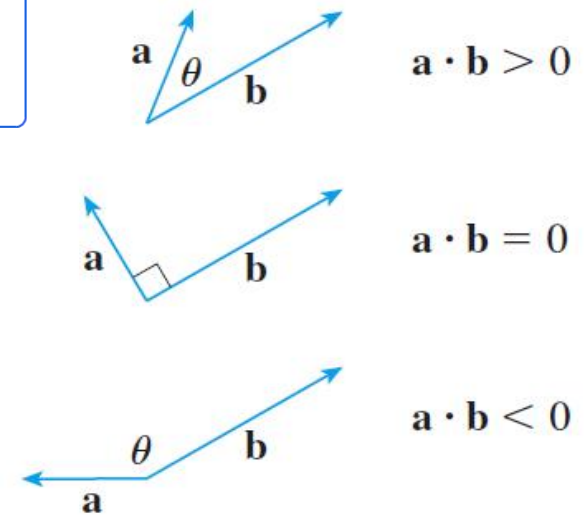
$$\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + a_2 b_2 + a_3 b_3$$

Theorem 1.2: If θ is the angle between the vectors $\mathbf{a}$ and $\mathbf{b}$, then

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| |\mathbf{b}| \cos \theta$$

Theorem 1.3:

Two vectors $\mathbf{a}$ and $\mathbf{b}$ are **orthogonal** if and only if $\mathbf{a} \cdot \mathbf{b} = 0$.



James Stewart, Calculus (6th Edition), Brooks/Cole Pub Co, 2007

1.3 The Trigonometric System and Orthogonality (三角法システムと直交性)

Orthogonality properties of the trigonometric system are expressed by

$$\int_{-\pi}^{\pi} \cos mx \cos nx \, dx = 0 \quad \text{if } m \neq n \quad (1.4)$$

$$\int_{-\pi}^{\pi} \cos mx \sin nx \, dx = 0 \quad \text{for all } m \text{ and } n \quad (1.5)$$

$$\int_{-\pi}^{\pi} \sin mx \sin nx \, dx = 0 \quad \text{if } m \neq n \quad (1.6)$$

For the case $n = m$, we have

$$\int_{-\pi}^{\pi} \cos^2 mx = \int_{-\pi}^{\pi} \sin^2 mx = \pi \quad \text{for all } m \neq 0 \quad (1.7)$$

1.3 The Trigonometric System and Orthogonality (三角法システムと直交性)

$$\int_{-\pi}^{\pi} \cos mx \cos nx \, dx = 0 \quad \text{if } m \neq n$$

Proof:

Use a [trigonometric identity \(Product-to-sum identities\)](#) and write

$$\cos mx \cos nx = \frac{1}{2} (\cos (m+n)x + \cos (m-n)x)$$

Since $m \pm n \neq 0$, we get

$$\begin{aligned} \int_{-\pi}^{\pi} \cos mx \cos nx \, dx &= \int_{-\pi}^{\pi} \left[\frac{1}{2} (\cos (m+n)x + \cos (m-n)x) \right] dx \\ &= \frac{1}{2} \left[\frac{1}{m+n} \sin (m+n)x + \frac{1}{m-n} \sin (m-n)x \right]_{-\pi}^{\pi} = 0 \quad \blacksquare \end{aligned}$$

1.4 Fourier Series

フーリエ級数

1.4 Fourier Series (フーリエ級数)

Fourier series are special expansions of 2π -periodic functions of the form

$$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) \quad (1.8)$$

Q: If a function has a Fourier series, how do we compute the coefficients $a_0, a_1, a_2, \dots, b_1, b_2, \dots$?

1.4 Fourier Series (フーリエ級数)

Euler Formulas for the Fourier Coefficients

We integrate both sides of (1.8) over the interval $[-\pi, \pi]$

Assuming term-by-term integration is justified, and get

$$\int_{-\pi}^{\pi} f(x) dx = \int_{-\pi}^{\pi} a_0 dx + \sum_{n=1}^{\infty} \int_{-\pi}^{\pi} (a_n \cos nx + b_n \sin nx) dx$$

But because

$$\int_{-\pi}^{\pi} \cos nx dx = \int_{-\pi}^{\pi} \sin nx dx = 0 \text{ for } n = 1, 2, \dots,$$

it follows that

$$\int_{-\pi}^{\pi} f(x) dx = \int_{-\pi}^{\pi} a_0 dx = 2\pi a_0, \quad \Rightarrow \quad \underline{a_0} = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx.$$

Note: a_0 is the average of f on the interval $[-\pi, \pi]$.

1.4 Fourier Series (フーリエ級数)

starting with (1.8), we multiply both sides by $\cos mx$ ($m \geq 1$), integrate term-by-term, use the orthogonality

$$\begin{aligned}\int_{-\pi}^{\pi} f(x) \cos mx \, dx &= \overbrace{\int_{-\pi}^{\pi} a_0 \cos mx \, dx}^{=0} + \sum_{n=1}^{\infty} \overbrace{\int_{-\pi}^{\pi} a_n \cos nx \cos mx \, dx}^{=0 \text{ for } m \neq n} \\ &\quad + \sum_{n=1}^{\infty} \overbrace{\int_{-\pi}^{\pi} b_n \sin nx \cos mx \, dx}^{=0} \\ &= a_m \overbrace{\int_{-\pi}^{\pi} \cos^2 mx \, dx}^{=\pi} = \pi a_m.\end{aligned}$$

Hence

$$\underline{a_m} = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos mx \, dx \quad (m = 1, 2, \dots).$$

1.4 Fourier Series (フーリエ級数)

By a similar procedure, starting with (1.8), we multiply both sides by $\sin mx$ ($m \geq 1$), integrate term-by-term, use the orthogonality

$$\underline{b_m} = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin mx \, dx \quad (m = 1, 2, \dots).$$

1.4 Fourier Series (フーリエ級数)

Euler Formulas for the Fourier Coefficients (フーリエ係数)

Suppose that f has the Fourier series representation

$$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) \quad (1.8)$$

Then the coefficients a_0 , a_n , and b_n are called the Fourier coefficients of f and are given by the following Euler formulas:

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx \quad (1.9)$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx \quad (n = 1, 2, \dots) \quad (1.10)$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx \quad (n = 1, 2, \dots) \quad (1.11)$$

1.4 Fourier Series (フーリエ級数)

Euler Formulas for the Fourier Coefficients (フーリエ係数)

Alternative Euler Formulas

$$a_0 = \frac{1}{2\pi} \int_0^{2\pi} f(x) dx \quad (1.12)$$

$$a_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos nx dx \quad (n = 1, 2, \dots) \quad (1.13)$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin nx dx \quad (n = 1, 2, \dots) \quad (1.14)$$

1.4 Fourier Series (フーリエ級数)

Example 1.3 Fourier series of the sawtooth function (のこぎり波関数)

The sawtooth function, shown in Figure 1.7, is determined by the formulas

$$f(x) = \begin{cases} \frac{1}{2}(\pi - x) & \text{if } 0 < x \leq 2\pi \\ f(x + 2\pi) & \text{otherwise} \end{cases}$$

Find its Fourier series.

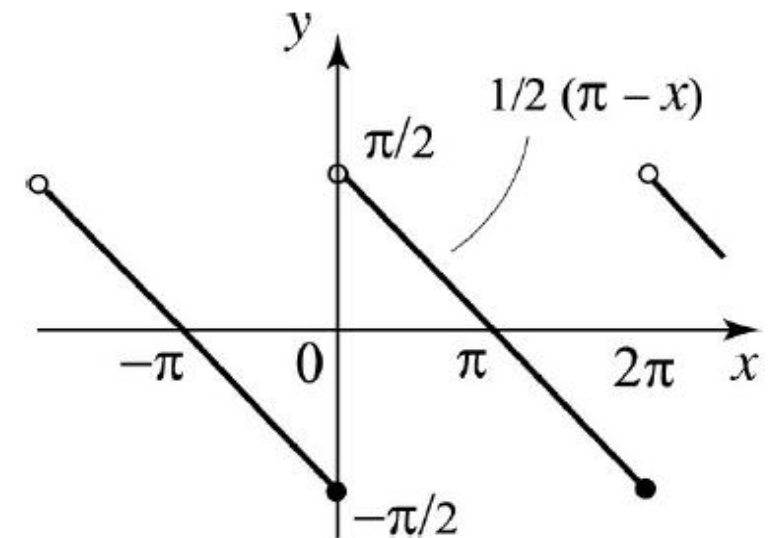


Figure 1.7 Sawtooth function

1.4 Fourier Series (フーリエ級数)

Solution (notes)

Solution: By using (1.12), (1.13), (1.14), we have

$$\begin{aligned}a_0 &= \frac{1}{2\pi} \int_0^{2\pi} f(x) dx = \frac{1}{2\pi} \int_0^{2\pi} \frac{1}{2} (\pi - x) dx \\&= \frac{1}{2\pi} \left[\frac{1}{2} \pi x - \frac{1}{4} x^2 \right]_0^{2\pi} \\&= \frac{1}{2\pi} \left[(\pi^2 - \frac{4}{4} \pi^2) - (0 - 0) \right] \\&= 0\end{aligned}$$

$$\begin{aligned}a_n &= \frac{1}{\pi} \int_0^{2\pi} \frac{1}{2} (\pi - x) \cos nx dx \\&= \frac{1}{2\pi} \left[\int_0^{2\pi} \pi \cos nx dx - \int_0^{2\pi} x \cos nx dx \right] \\&= \frac{1}{2\pi} \left[\int_0^{2\pi} \frac{1}{n} \pi \cos nx d(nx) - \frac{1}{n^2} \int_0^{2n\pi} nx \cdot \cos nx \cdot d(nx) \right] \text{ (change } x \text{ to } nx)\end{aligned}$$

$$\begin{aligned}&= \frac{1}{2\pi} \left[\frac{\pi}{n} \sin nx \Big|_0^{2n\pi} - \frac{1}{n^2} \int_0^{2n\pi} nx \cos nx d(nx) \right] \\&= \frac{1}{2\pi} \left[0 - \left(\frac{1}{n^2} nx \cdot \sin nx \Big|_0^{2n\pi} - \frac{1}{n^2} \int_0^{2n\pi} \sin nx d(nx) \right) \right] \\&= -\frac{1}{2\pi} \left[0 - \frac{1}{n^2} \cos nx \Big|_0^{2n\pi} \right] \\&= \frac{1}{2\pi} \left[\frac{1}{n^2} \cos(n \cdot 2n\pi) - \frac{1}{n^2} \cos(n \cdot 0) \right] \\&= \frac{1}{2\pi} [1 - 1] \\&= 0\end{aligned}$$

integration by parts

$$u = nx \quad du = d(nx)$$

$$dv = \cos(nx) d(nx)$$

$$v = \sin(nx)$$

$$\frac{1}{n^2} \left\{ [u \cdot v]_0^{2n\pi} - \int_0^{2n\pi} v du \right\}$$

$$= \frac{1}{n^2} nx \cdot \sin(nx) \Big|_0^{2n\pi} - \frac{1}{n^2} \int_0^{2n\pi} \sin(nx) d(nx)$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} \frac{1}{2} (\pi - x) \sin nx dx$$

$$= \frac{1}{n} \quad (\text{similar to computing } a_n)$$

1.4 Fourier Series (フーリエ級数)

In evaluating a_n , we use the formula $\int x \cos nx \, dx = \frac{1}{n^2} \cos nx + \frac{x}{n} \sin nx$ which is obtained by integration by parts[†].

Substituting these values for a_n and b_n into (1.8), we obtain $\sum_{n=1}^{\infty} \frac{\sin nx}{n}$ as the **Fourier series of f** .

[†]Integration by parts (部分積分):

(English) https://en.wikipedia.org/wiki/Integration_by_parts

(Japanese) <https://ja.wikipedia.org/wiki/%E9%83%A8%E5%88%86%E7%A9%8D%E5%88%86>

1.4 Fourier Series (フーリエ級数)

Integration by parts (部分積分)

Product of two functions [edit]

The theorem can be derived as follows. Suppose $u(x)$ and $v(x)$ are two [continuously differentiable functions](#). The [product rule](#) states (in [Leibniz's notation](#)):

$$\frac{d}{dx} (u(x)v(x)) = v(x) \frac{d}{dx} (u(x)) + u(x) \frac{d}{dx} (v(x)).$$

Integrating both sides with respect to x ,

$$\int \frac{d}{dx} (u(x)v(x)) dx = \int u'(x)v(x) dx + \int u(x)v'(x) dx$$

then applying the definition of [indefinite integral](#),

$$u(x)v(x) = \int u'(x)v(x) dx + \int u(x)v'(x) dx$$

$$\int u(x)v'(x) dx = u(x)v(x) - \int u'(x)v(x) dx$$

gives the formula for **integration by parts**.

Since du and dv are [differentials of a function](#) of one variable x ,

$$du = u'(x)dx \quad dv = v'(x)dx$$

$$\int u(x) dv = u(x)v(x) - \int v(x) du$$

The original integral $\int uv' dx$ contains v' ([derivative](#) of v); in order to apply the theorem, v ([antiderivative](#) of v') must be found, and then the resulting integral $\int vu' dx$ must be evaluated.

Polynomials and trigonometric functions [edit]

In order to calculate

$$I = \int x \cos(x) dx ,$$

let:

$$u = x \Rightarrow du = dx$$

$$dv = \cos(x) dx \Rightarrow v = \int \cos(x) dx = \sin(x)$$

then:

$$\begin{aligned} \int x \cos(x) dx &= \int u dv \\ &= u \cdot v - \int v du \\ &= x \sin(x) - \int \sin(x) dx \\ &= x \sin(x) + \cos(x) + C, \end{aligned}$$

where C is a [constant of integration](#).

For higher powers of x in the form

$$\int x^n e^x dx, \quad \int x^n \sin(x) dx, \quad \int x^n \cos(x) dx ,$$

repeatedly using integration by parts can evaluate integrals such as these; each application of the theorem lowers the power of x by one.

1.4 Fourier Series (フーリエ級数)

Note 1: At the points of discontinuity ($x = 2k\pi$) in Example 1, the Fourier series converges to 0 which is the average value of the function from the left and the right at these points.

Note 2: Near the points of discontinuity, the Fourier series overshoots its limiting values, known as the Gibbs phenomenon.

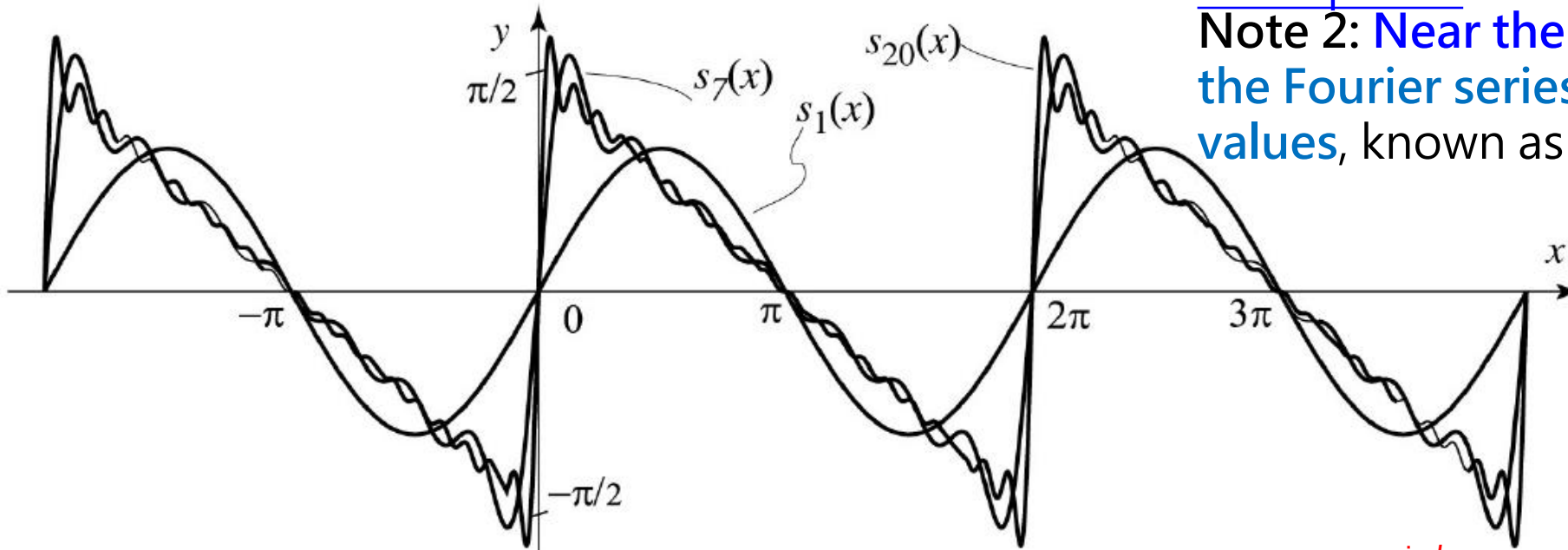


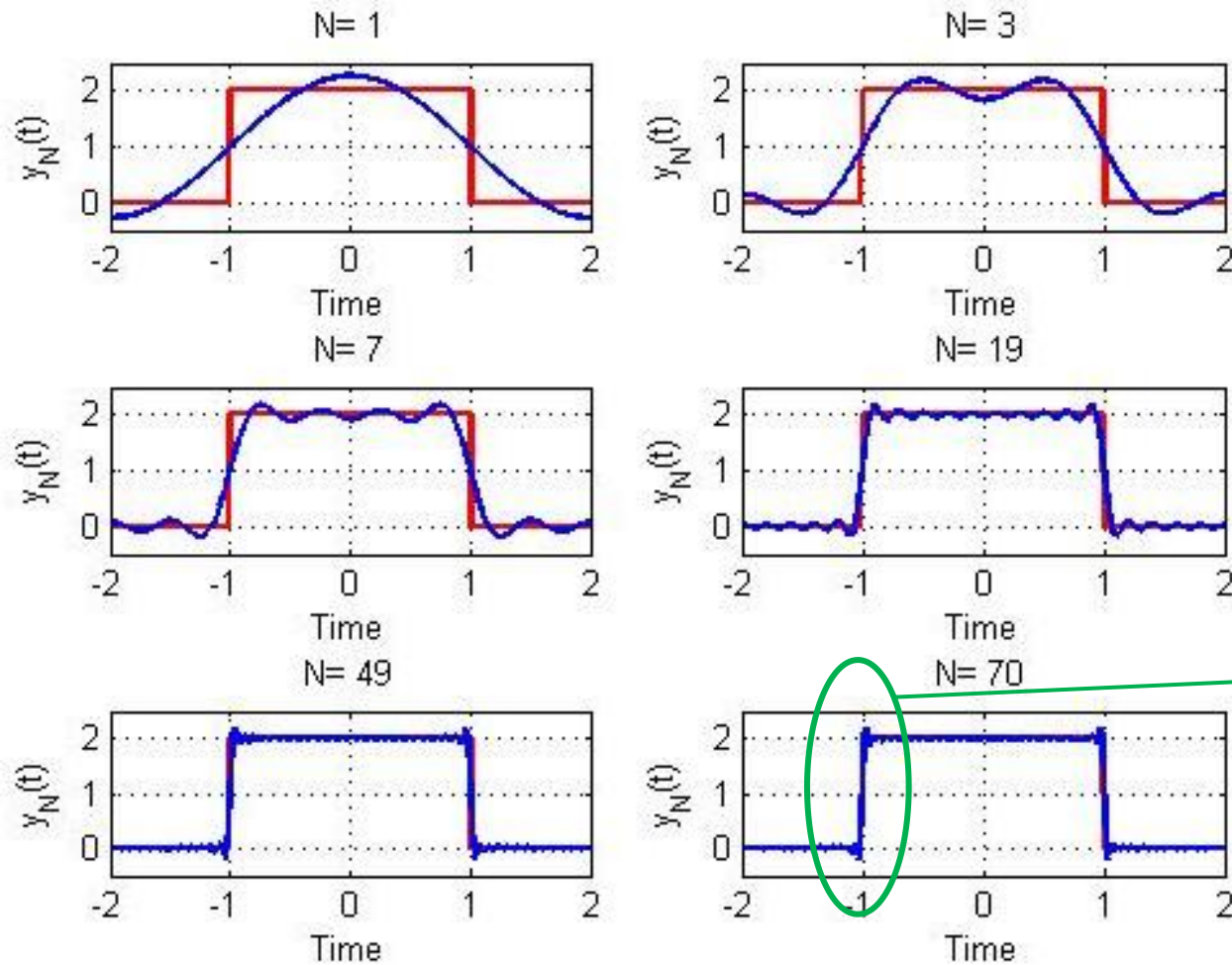
Figure 1.8 Here the n th partial sum of the Fourier series is $s_n(x) = \sum_{k=1}^n \frac{\sin kx}{k}$.

Distinguish the graphs, note that as n increases, the frequencies of the sine terms increase.

This causes the graphs of the higher partial sums to be more wiggly. The limiting graph is the graph of the whole Fourier series, shown in Figure 1.9. It is identical to the graph of the function, except at points of discontinuity.

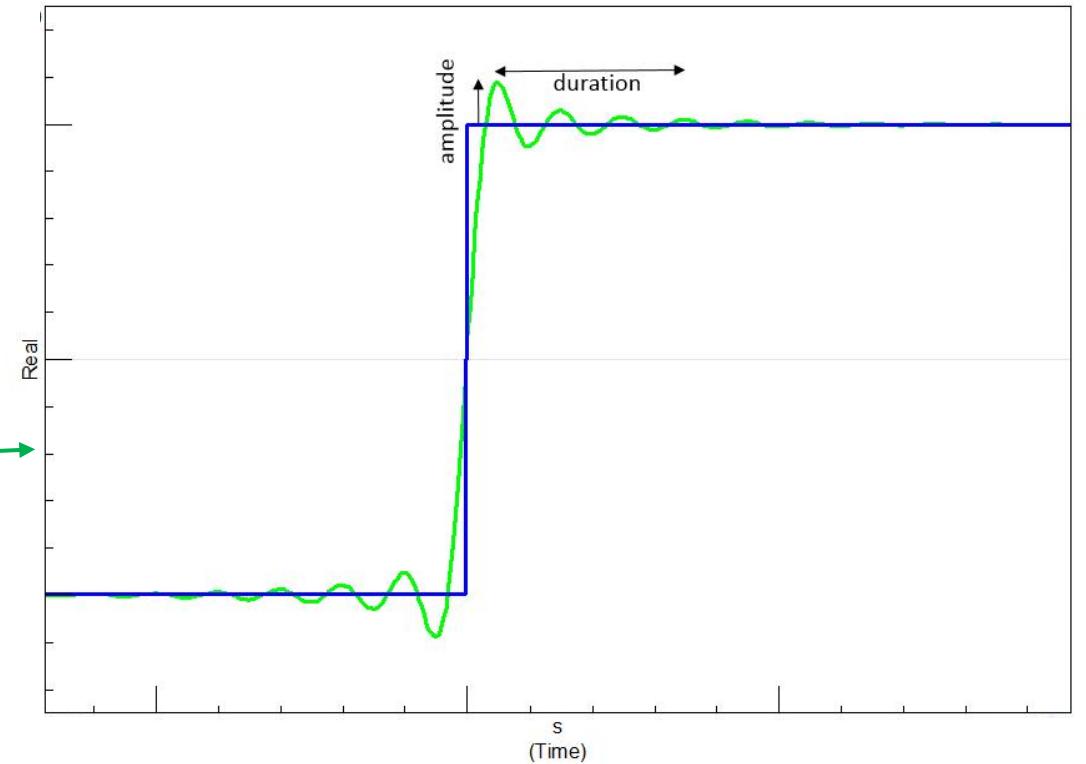
1.4 Fourier Series (フーリエ級数)

Gibbs phenomenon ギブズ現象



Gibbs phenomenon† is the peculiar manner in which the Fourier series of a piecewise continuously differentiable periodic function behaves at a jump discontinuity.

ギブズ現象‡は、区分的連続微分可能な周期関数のフーリエ級数において、その関数が第1種不連続 (discontinuity of the first kind 又は jump discontinuity) となる点付近では、フーリエ級数の n 次部分和が大きく振動して、部分和の最大値が関数自体の最大値より大きくなってしまいうことがあるという振る舞いのことを指す。



† Gibbs phenomenon https://en.wikipedia.org/wiki/Gibbs_phenomenon

‡ ギブズ現象 <https://ja.wikipedia.org/wiki/%E3%82%AE%E3%83%96%E3%82%BA%E7%8F%BE%E8%B1%A1>

Review for Lecture 1

- Periodic Functions
- Piecewise Continuous Functions
- Piecewise Smooth Functions
- The Trigonometric System and Orthogonality
- Fourier Series

Assignment

Please Check <https://web-ext.u-aizu.ac.jp/~xiangli/teaching/MA05/index.html>

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